Chinese-SkillSpan: a benchmark for competency span extraction in Chinese job advertisements

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ABSTRACT

Competency spans in Chinese job advertisements are a useful source for labour market research. The problem is annotation: it is hard to do consistently. Chinese has no spaces between words. Annotators therefore have to rely on meaning and context to determine span boundaries. The same term may also work as a skill in one clause and as a knowledge requirement in another. Existing resources also follow different annotation schemes. We present Chinese-SkillSpan, a benchmark of source-text spans with character-level offsets, built from recruitment advertisements from several sources. We also release the handbook behind it, which was produced through human annotation, disagreement review, and adjudication. Inspired by the high-level categories of the European Skills, Competences, Qualifications and Occupations (ESCO) classification, it sorts spans into four categories: language, knowledge-related requirements, occupational skills, and transversal competencies. Supervised training uses machine-generated "silver" labels and excludes the human-reference sentences. Silver labels were checked with automated tests, and a sample was reviewed by hand. Prompted language models, the Chinese recruitment model JobBERT-zh + CRF, multilingual encoders and silver-adapted Qwen were evaluated on the same human reference set with shared scoring rules. A prompted GPT model ranked highest among evaluated configurations, without a clear advantage over the next-highest configuration. Qwen adapted with silver labels performed better than the unadapted model under the same inference settings. In a separate small blinded study, boundary-only agreement was slightly higher than typed agreement. Researchers can use Chinese-SkillSpan to compare models and analyse boundary and type errors.

INTRODUCTION

Job markets keep changing, and so do the requirements that employers write into their advertisements (Podjanin and Strietska-Ilina, 2020; Senger et al., 2024). Advertisements record

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which competencies are asked for, and extracting them from unstructured text is one way to make that record usable. Span extraction does this by marking requirement mentions in the source text, so a reader can check where each span begins and ends and what type it received (Gnehm et al., 2022). Document-level methods take a different route and assign skills from a predefined inventory (Bhola et al., 2020).

Existing resources differ in annotation schemes and prediction targets (Table 1). English SkillSpan permits overlapping SKILL and KNOWLEDGE layers (Zhang et al., 2022b). Kompetencer adds fine-grained labels derived from the European Skills, Competences, Qualifications and Occupations (ESCO) classification to annotated Danish and English spans through distant supervision (Zhang et al., 2022a), and work on German-language job advertisements links requirement spans to taxonomy concepts (Gnehm et al., 2022). Chinese recruitment studies cover skill and specialty annotation (Cao et al., 2021) and skill-demand forecasting (Chen et al., 2024). Chinese adds further difficulties. The text has no spaces between words, so tokenisation alone cannot show an annotator where a competency span begins and ends (Peng and Dredze, 2015; Chen and Kong, 2021). An action must stay with its objects whenever splitting them would change the requirement. Explicit, context-sensitive conventions for Chinese boundaries and types are therefore needed for consistent annotation.

We introduce Chinese-SkillSpan, a benchmark of source-text spans with character-level offsets (Figure 1). We use *competency span* for any contiguous mention of a requirement within the handbook's scope in the source text, and reserve *occupational skills* for one of its types. The handbook assigns each span to one of four types: language (L), knowledge-related requirements (K), occupational skills (S), or transversal competencies (T). The types draw on ESCO's high-level categories and on the knowledge–skill distinction in the European Qualifications Framework (European Commission, 2024; Council of the European Union, 2017), but the scheme assigns no ESCO concept identifiers. Human reference labels for evaluation are kept separate from machine-generated silver labels for supervision. Table 2 gives handbook examples.

We compare prompted language models, the Chinese recruitment model JobBERT-zh + CRF, a Qwen model adapted with silver labels, and multilingual encoders under a shared evaluation protocol. Under fixed inference settings, silver-label adaptation improves end-to-end extraction over the same model without adaptation. Among the configurations evaluated, a prompted GPT model ranks highest, without a clear advantage over the next-highest configuration. The human reference set was selected from sentences where model annotations disagreed, so it concentrates on ambiguous cases that came up during handbook development. In a separate small blinded study, agreement on span boundaries alone was slightly higher than agreement that also required types. We document the annotation layers, evaluation targets, and procedures, following guidance on dataset documentation and annotation agreement (Bender and Friedman, 2018; Gebru et al., 2021; Artstein and Poesio, 2008).

An earlier preprint gave a preliminary account of Chinese-SkillSpan (Li et al., 2026); this article substantially revises its resource definitions, annotation procedures, and evaluation, and Appendix D separates the earlier reports from the experiments reported here.

This study makes three contributions. First, it provides a Chinese competency-span resource with character-level offsets, and it keeps human reference and silver annotation layers separate. Second, it develops a handbook with context-sensitive boundary and type rules, illustrated with

positive and negative examples. Third, it reports annotation-quality assessments and model baselines under shared scoring procedures, shows that silver-label adaptation improves extraction, and analyses the boundary and type errors that models make.

RELATED WORK

Recruitment extraction resources. Recruitment datasets differ in their prediction units and annotation targets. Candidate-based classification determines whether a supplied soft-skill phrase refers to an applicant (Sayfullina et al., 2018), while document-level retrieval assigns skills from a predefined inventory to an advertisement (Bhola et al., 2020). Span-based resources preserve the wording of individual requirements. SkillSpan provides English annotations in potentially overlapping SKILL and KNOWLEDGE layers, covering both hard and soft skills (Zhang et al., 2022b). Kompetencer extends Danish and English skill and knowledge spans with ESCO-derived fine-grained labels (Zhang et al., 2022a). For German advertisements, Gnehm et al. (2022) combine requirement-span extraction with finer-grained skill areas and taxonomy matching.

Chinese recruitment research includes entity annotation for occupational profiling and benchmarks for skill-demand forecasting. Cao et al. (2021) manually annotated skill and specialty requirements and trained a BERT–BiLSTM–CRF model, while Job-SDF evaluates forecasting at several aggregation levels (Chen et al., 2024). Chinese-SkillSpan focuses on extracting source-text requirements as flat character-level spans with four competency types. Table 1 compares resources side by side: prediction units, annotation schemes, and available research materials.

Competency annotation and guidelines. Competency annotation depends on how boundaries and types are defined in context. Chinese named entity recognition studies address social-media text (Peng and Dredze, 2015), fine-grained categories (Xu et al., 2020), and boundary detection without explicit word delimiters (Chen and Kong, 2021). Multilingual benchmarks also examine noisy text and context-dependent entity types (Fetahu et al., 2023; Luo et al., 2025). In recruitment text, coordinated and ambiguous mentions create annotation difficulties across languages (Nguyen et al., 2024), while variation in terminology and limited public annotations complicate comparisons among resources (Senger et al., 2024).

ESCO provides a multilingual vocabulary for labour-market information (le Vrang et al., 2014; European Commission, 2024), and SkiLLens combines human involvement with skill extraction and ESCO linking for European job advertisements (De Santo et al., 2026). Chinese-SkillSpan uses ESCO's high-level distinctions to guide Chinese span annotation. Its annotation rules specify both span boundaries and category assignment (Artstein and Poesio, 2008).

Model-assisted annotation and extraction. Weak supervision and model assistance offer ways to extend manually annotated resources. Taxonomy-based methods derive labels from skill inventories (Zhang et al., 2022c), while large language models (LLMs) support automated and collaborative annotation (Ding et al., 2023; Li et al., 2023). Ul Haq et al. (2026) evaluate LLM-generated labels and their utility for downstream model training, including on SkillSpan. GoLLIE uses annotation guidelines for information extraction (Sainz et al., 2024), and Kim et al. (2026) study guideline refinement for LLM annotation in biomedical NER. JobSkape generates synthetic job advertisements for skill-to-taxonomy matching (Magron et al., 2024). Our silver labels for collected recruitment texts undergo structural checks and sampled human review.

Extraction models draw on both general language representations and recruitment-specific

Table 1. Recruitment resources: prediction units, annotation schemes, and materials.

Study / language	Prediction unit	Annotation scheme and label source	Reported materials
Sayfullina et al. (2018); English	Candidate phrases in context	Binary labels indicating whether a supplied soft-skill phrase refers to an applicant	Annotated soft-skill contexts
Gnehm et al. (2022); German	Requirement spans and finer-grained skill areas	Coarse EDU/EXP/LNG labels; finer-grained skill areas and taxonomy matching	Article and released domain-adapted models
SkillSpan (Zhang et al., 2022b); English	Source-text spans	SKILL and KNOWLEDGE in two potentially overlapping annotation layers	Dataset, code, and annotation guidance
Kompetencer (Zhang et al., 2022a); Danish / English	Spans and fine-grained labels	Manually annotated skill/knowledge spans; distant labels from the ESCO API; manually corrected Danish test labels	Dataset and code
Cao et al. (2021); Chinese	Recruitment entities	Manually annotated skill and specialty entities encoded with BIOES	Article and supporting data files
Chinese-SkillSpan; Chinese	Flat character-level spans	L/K/S/T types; human reference labels and model-generated silver supervision	Core dataset, handbook, protocols, baseline predictions, scoring code, and individual blinded-annotation layers

Notes. EDU, EXP, and LNG denote education, experience, and language requirements, respectively.

information. Pretrained encoders and structured decoding provide a basis for sequence labelling (Devlin et al., 2019; Lafferty et al., 2001), while continued pretraining adapts representations to a target domain (Gururangan et al., 2020). SkillSpan applies this approach to recruitment text through JobBERT and JobSpanBERT (Zhang et al., 2022b). ESCOXLM-R incorporates taxonomy information during multilingual pretraining (Zhang et al., 2023), and NNOSE uses retrieval over training-data representations to support skill extraction (Zhang et al., 2024).

General information extraction methods offer additional approaches to span prediction. UIE uses text-to-structure generation (Lu et al., 2022), UniversalNER learns from distilled supervision (Zhou et al., 2024), and GLiNER supports open-type extraction with an encoder (Zaratiana et al., 2024). GPT-NER formulates named entity recognition as generative extraction with self-verification (Wang et al., 2025). We compare prompted extraction, silver-supervised adaptation, and encoder-based prediction for Chinese competency spans.

DATASET CONSTRUCTION AND ANNOTATION

Chinese-SkillSpan combines a Chinese recruitment corpus, an annotation handbook, human reference labels for evaluation, and silver labels for supervision. Figure 1 shows how corpus preparation and annotation support the model comparisons.

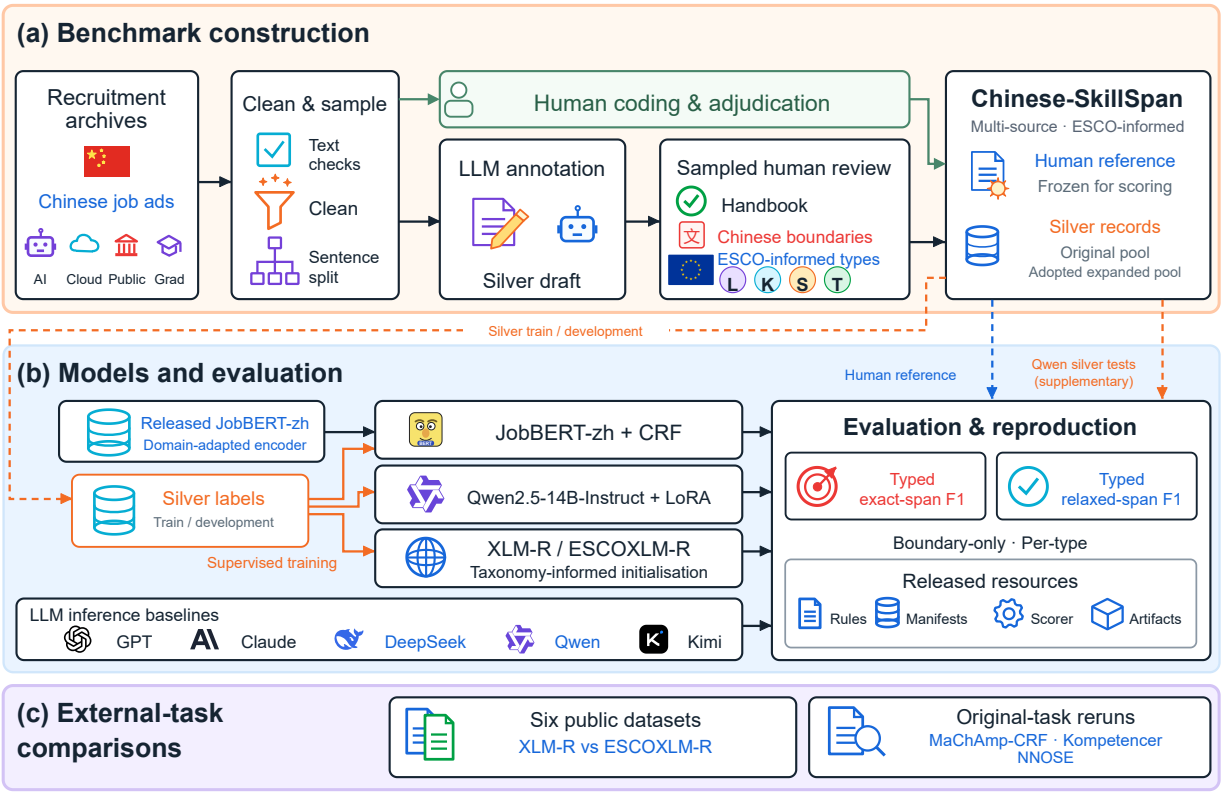


Figure 1. Overview of Chinese-SkillSpan: (a) benchmark construction; (b) models and evaluation; and (c) external-task comparisons.

Competency Definitions and Annotation Schema

Competency research connects people's capabilities with the activities they perform. McClelland (1973) argued for assessing competence through behaviour relevant to meaningful performance criteria. ESCO defines competence in terms of demonstrated use of knowledge, skills, and personal, social, or methodological abilities in work and learning, with responsibility and autonomy (European Commission, nda). Here, we use *competency* as an umbrella term for capabilities and associated requirements expressed in Chinese recruitment text.

ESCO offers a cross-occupation vocabulary for labour-market requirements. Its skills pillar distinguishes knowledge concepts from skill/competence concepts, while grouping skills and competences together (European Commission, ndb). The European Qualifications Framework defines knowledge through what is learned about a field, including its facts, principles, theories, and practices. Skills concern putting knowledge and practical know-how to use in performing tasks and addressing problems (Council of the European Union, 2017). This distinction also informs SkillSpan's skill and knowledge annotations (Zhang et al., 2022b).

Chinese-SkillSpan uses ESCO as a conceptual basis for four task-specific annotation categories (European Commission, 2024). Language requirements receive L, knowledge-related requirements K, occupational skills S, and transversal competencies T. L separates natural-language requirements, while T covers broadly applicable communication, cooperation, self-management, and related dispositions. The project-specific K category includes qualifications

and explicit industry-background requirements alongside substantive knowledge. Table 2 summarises the categories with bilingual examples; the handbook guides Chinese span annotation.

Given a Chinese job-advertisement sentence x, the task is to predict a flat set of non-overlapping spans (s, e, t), where $t \in \{L, K, S, T\}$. Offsets are measured in Unicode code points: s denotes the zero-based start position, and e denotes the exclusive end position. Each mention must exactly match the contiguous substring $x[s : e]$, preserving spelling and internal spaces.

Table 2. Operational L/K/S/T definitions and Chinese span examples (handbook v4.2.14).

Label	Operational definition	Chinese examples and English glosses
L: Language	Natural-language names, proficiency levels, examinations, and certificates.	英语六级 *College English Test Band 6* 日语 N2 *Japanese proficiency at N2 level*
K: Knowledge-related requirements	Facts, principles, theories, and fields of study; educational and non-language qualifications; and explicit industry-background requirements.	SQL 原理 *Principles of SQL* 本科及以上学历 *Bachelor's degree or higher*
S: Occupational skill	Occupational activities and the application of tools or methods.	使用 [SQL] *Use [SQL]* 数据分析 *Data analysis*
T: Transversal competency	General communication, cooperation, cognition, self-management, and dispositions.	客户汇报 *Reporting to clients* 责任心 *Sense of responsibility*

Data Sources and Annotation Layers

The source corpus contains 22,840 sentences from four collection batches: AI, Cloud, public-sector, and graduate recruitment. Texts came from commercial collections purchased from MacroData (MacroData, nd), public-institution announcements collected by the research team, and the Recruitment Dataset on Alibaba Cloud Tianchi (Tianchi, nd). For public-institution texts, the team selected government, university, hospital, and public recruitment websites across China, then screened and cleaned their texts. The batches may cover overlapping occupations.

Table 3 maps the principal annotation layers to their roles, following guidance on dataset documentation (Bender and Friedman, 2018; Gebru et al., 2021). The human reference provides evaluation labels, while records selected from the original silver pool provide training and development labels. The adopted expanded silver pool combines earlier supervision with additional cloud, public-sector, and listed-company recruitment texts selected using requirement-related keywords. The model-specific training and development splits are described in Experimental Setup.

The human reference contains 663 spans: 448 S, 125 K, 88 T, and two L. The 150 sentences have distinct complete texts after Unicode Normalization Form C (NFC). This operation gives

Table 3. Source corpus and principal annotation layers in Chinese-SkillSpan.

Subset	Size	Annotation content	Role
Source corpus	22,840 sentences	Recruitment text from four collection batches	Source material for annotation
Human reference	150 sentences	663 spans; 8 sentences with no spans (5.3%)	Model evaluation
Original silver pool	2,451 records	Model-generated annotations with sampled human review	Candidate supervision
Adopted expanded silver	9,540 records	21,101 spans; 2,492 records with no spans (26.1%)	Expanded supervision and evaluation against silver targets

Pool sizes count annotation records, which may repeat a sentence. The layers overlap in source material and should not be summed.

equivalent character sequences a consistent representation for comparison. An alternative expanded pool of 9,646 records supports a supplementary annotation-channel comparison. Appendix D details both pools' source partitions, composition, and selection criteria.

Annotation Guidelines and Adjudication

The handbook answers three questions: what counts as a requirement, where its span begins and ends in the source text, and which type label applies given the sentence context. Scope is assessed clause by clause: benefits, application procedures, company promotion, and outcomes alone are excluded, while recruitment activities can qualify as occupational duties. A confirmed absence of in-scope requirements receives an empty annotation. Missing context or unresolved interpretation is handled separately, including in sentences that mix requirements with other content.

Boundary selection prioritises semantic completeness. Following SkillSpan (Zhang et al., 2022b), annotators retain the shortest complete source span, with no fixed length limit or added words. Generic proficiency triggers are usually excluded; modal verbs, heads, and complements remain when needed to express the capability. Coordinated items are split only if each is meaningful in context; shared actions retain their objects when needed to preserve meaning.

Type assignment uses the requirement's context. Using SQL supports S for the tool mention, whereas understanding the principles of SQL supports K for the complete knowledge phrase. General communication and coordination receive T even when stated as duties, while technical operations receive S. Experience expressions follow the same contextual approach: generic duration alone is excluded, explicit industry-background requirements receive K, and an experience qualifier is retained when its removal would change the requirement.

Adjudication resolves scope, boundaries, and type separately (Artstein and Poesio, 2008). Records with unresolved annotations are excluded from labelled training; alternative or overlapping spans remain separate from final non-overlapping annotations. Figure 2 pairs human-reference annotations with English translations; Appendix A gives rules G1–G15 and examples.

The examples illustrate distinct type and boundary decisions. In Figure 2a, English reading and writing receive L, while communication and teamwork receive T. In panel (b), object orientation and design patterns receive K, while multithreaded and cross-platform programming forms one S span. Panel (c) separates occupational actions into S spans and retains exposure and conversion



Figure 2. Chinese human-reference spans (metadata: v4.2.10) with English glosses for readers. Colours and letters align spans; ellipses mark omitted context. Source sentence IDs and character offsets are recorded in the linked example metadata.

rates together with their shared optimisation action. List numbers are omitted from the illustration; the original texts and offsets are preserved in the data.

Annotation Process and Quality Assessment

The human reference set was selected from sentences flagged for disagreement between model annotations, to examine ambiguous requirements. The 50 calibration sentences were annotated separately and subsequently adjudicated. The authors recall that machine suggestions were available, whereas an earlier protocol specifies raw-text-only coding. The available records do not resolve this discrepancy, so calibration is not treated as a blinded agreement study (Appendix B). A separate 100-sentence challenge sample was collaboratively annotated and reviewed. Both samples informed handbook development, and their finalised labels form the 150-sentence reference used for model evaluation (Figure 3).

Silver labels were generated with GPT-6 Astra through OpenAI Codex, following the handbook and an initial sample review. Automated checks assessed annotation structure and record

coverage, including span text, offsets, and labels; sampled human review assessed whether the annotations captured the requirements. Additional assisted annotation used suggestions from GPT-6 Astra, Grok, Qwen, DeepSeek, and Kimi (Appendix C). Human annotation and review cover 550 distinct sentences across the reference, silver-review samples, and blinded study (Appendix B).

The handbook was refined across annotation stages. All 150 human-reference records carry v4.2.10 metadata, while the reference documentation identifies v4.2.11 as the operating handbook. Table 2 illustrates later v4.2.14 rules; it does not replace the preserved reference labels used for scoring. Appendix A.4 identifies the versions associated with each study. Reported scores use the preserved labels, prompts, and evaluation inputs. Dated Chinese and English editions are available in the handbook index. The latest reviewed edition is intended for new annotation; reproduction requires the original study materials.

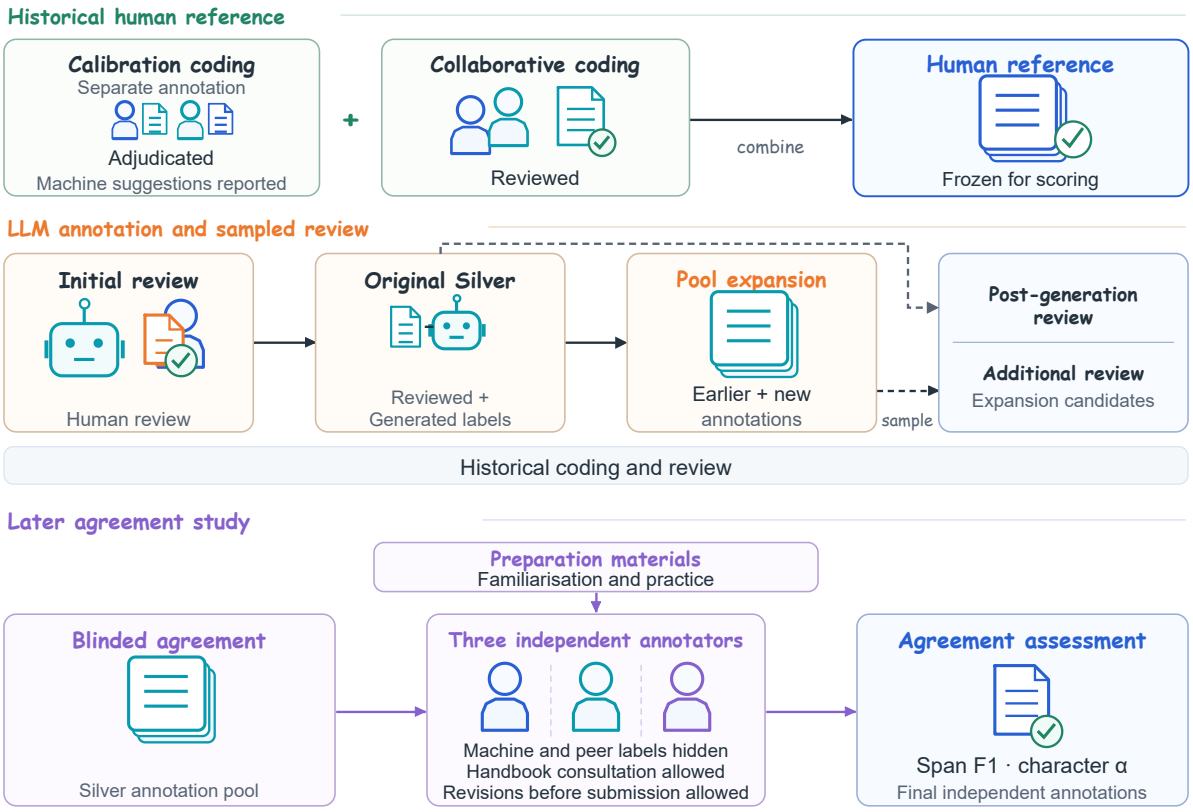


Figure 3. Human-reference construction, silver annotation and sampled review, and the later blinded agreement study. The calibration stage reflects the authors' reconstruction (Appendix B); machine and peer labels were hidden in the blinded study.

Two assisted-review studies assessed generated labels with machine suggestions visible. In a post-generation sample of 100 sentences, separate from the initial 100 reviewed records, agreement between machine-prefilled and human-reviewed annotations reached typed exact-span F1 of 0.959. Two reviewers separately annotated an additional 150-sentence sample, achieving pre-adjudication typed exact-span F1 of 0.816 (Appendix B).

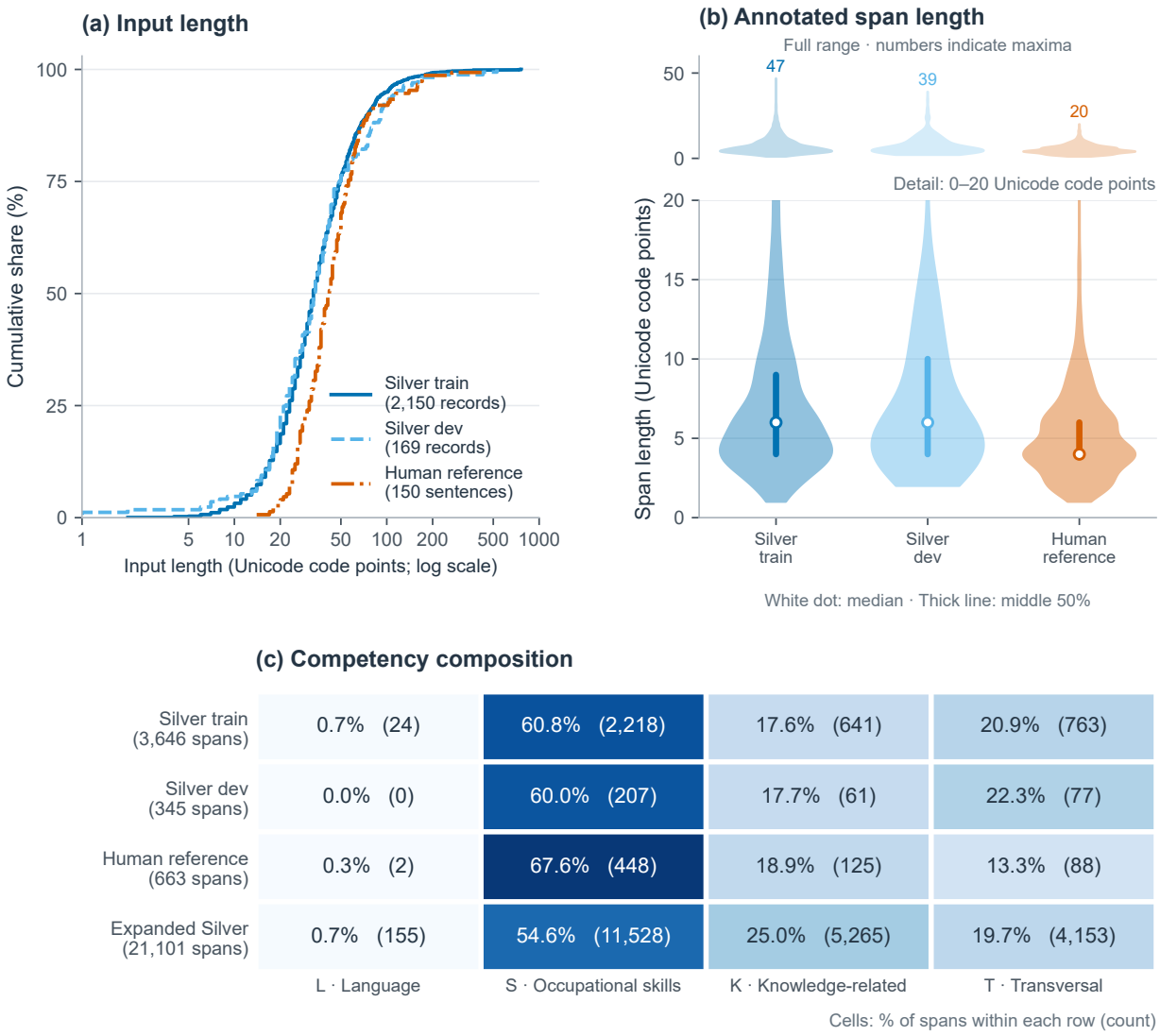


Figure 4. Dataset characteristics: (a) input lengths; (b) annotated span lengths; and (c) competency-type distributions, including the full adopted expanded pool. Silver train/dev denote the splits used by Qwen, XLM-R-large, and ESCOXLM-R. Lengths are measured in Unicode code points, including punctuation and spaces. The distributions include every record, including records whose source texts are identical.

Three annotators independently labelled the same 50-sentence sample drawn from sentences considered for silver annotation, with one sentence per advertisement identifier. They worked in separate Doccano projects with familiarisation and practice materials (Klie et al., 2024); machine suggestions and other annotators' labels were hidden throughout. Annotators could consult the handbook and revise their own labels before submission. The exact handbook file used for formal coding has not been established (Appendix A.4). Agreement was calculated from the final independent annotations, excluding familiarisation and practice sentences (Appendix B).

Table 4. Agreement on the 50-sentence blinded sample.

Measure	Estimate	95% interval	Agreement assessed
Mean pairwise typed exact-span F1	0.818	[0.732, 0.894]	Complete boundaries and types
Mean pairwise boundary-only exact-span F1	0.850	[0.766, 0.923]	Complete boundaries
Nominal Krippendorff's α	0.907	[0.851, 0.954]	L/K/S/T/O assignments across three annotators

Notes. Span F1 averages the three annotator-pair scores. Character agreement covers 2,111 Unicode code points. Intervals use a source-stratified sentence bootstrap.

Mean pairwise typed exact-span F1 was 0.818, compared with 0.850 when matching required only identical boundaries (Table 4). The lower typed score suggests additional disagreement over type assignment, although the confidence intervals overlap. Pairwise span F1 compares complete annotations symmetrically (Hripcsak and Rothschild, 2005), while character-level coefficients assess agreement at individual source positions (Artstein and Poesio, 2008). Across the three annotators, nominal Krippendorff's α was 0.907. The separate 50-sentence study does not provide an independent estimate of annotation reliability for the 150-sentence reference.

The adopted expanded silver pool labels 49 of the 50 blinded-study sentences. Of these 49 sentences, 46 are in training, two in validation, and one in test; the alternative pool has the same membership. The sample therefore does not provide an unseen test set for models trained on the expanded pools (Appendix D). A post hoc comparison with each annotator's labels on these sentences yielded mean typed exact-span micro-F1 of 0.709; Appendix B gives per-annotator results and compares the alternative pool.

Dataset Characteristics

The annotation layers differ in both type composition and span length (Figure 4). Occupational skills account for 54.6% of spans in the adopted expanded pool and 67.6% in the human reference. Language requirements are uncommon in both, accounting for less than 1% of spans. The median span length is four Unicode code points in the reference and six in the silver training and development sets used by Qwen, XLM-R-large, and ESCOXLM-R.

The proportion of records with no annotated span also differs: 40.4% in the silver training set and 48.5% in development, compared with 26.1% in the full expanded pool and 5.3% in the human reference. These differences motivate the analyses by type, span length, and presence of reference spans. Appendix B gives the split-level counts and additional-review distributions.

Ethics and Data Use

The study analysed recruitment advertisements, with annotation and review conducted by members of the research team. It did not recruit job applicants or collect applicant responses. Institutional ethics approval was not sought, and no formal exemption is claimed. The authors confirm permission to use the source materials for academic research and to share the MacroData materials for non-commercial academic use. Reuse remains subject to component-specific terms; Appendix D describes automated contact-pattern screening of already-public files and the limits of the documented privacy review.

EXPERIMENTAL SETUP

We compare prompted extraction, adaptation with silver supervision, and taxonomy-informed encoder initialisation using a common human reference set and shared scoring rules. The Qwen adaptation and encoder-initialisation comparisons hold their respective inference or training protocols fixed; supplementary experiments examine two expanded silver pools.

Data and Splits

The 150-sentence human reference set described in Dataset Construction and Annotation provides the evaluation labels for all main model comparisons. JobBERT-zh + CRF uses 2,156 silver training records and 169 development records. Qwen, XLM-R-large, and ESCOXLM-R use 2,150 training records and the same 169 development records after removing six training records whose complete texts matched three development texts after NFC normalisation.

In the 2,150/169 split, training and development share no complete sentences as stored or after NFC normalisation. A broader check using NFKC normalisation, whitespace removal, and case folding found two normalised complete-text keys shared by training and development, but none shared by training and the human reference. Human-reference sentences are excluded by identifier and normalised text. However, 59 of the 61 source-document groups identified from the reference record IDs also occur in the silver training set; these shared groups contain 143 of the 150 reference sentences (Appendix D).

Supplementary Qwen experiments use the adopted and alternative expanded silver pools. Each uses sentence-level training, validation, and test splits; shared records retain the same split assignments across pools. Validation labels determine checkpoint selection, and silver-test scores measure agreement with generated annotations. Both configurations are also evaluated on the common human reference, whose sentences are excluded from fitting and model selection. Each expanded pool has one normalised complete-text key shared by training and validation under the broader check (Appendix D). The pools differ in records and annotation channel; Appendix C lists their split sizes and training settings.

Models and Comparisons

Prompted inference compares GPT-5.6 Terra, Claude Sonnet 4.5, DeepSeek-V4-Pro, Kimi K2.6, and Qwen2.5-14B-Instruct (Yang et al., 2025) under shared annotation instructions. DeepSeek is evaluated with and without thinking, giving six prompted configurations. Table 5 lists access routes and inference settings. GPT and Claude are accessed through APIs; their names follow the service responses. The GPT-5.6 Terra evaluation configuration differs from GPT-6 Astra used through Codex for silver generation (Appendix C). Request and response identifiers are documented in the reproduction materials linked in Appendix C.

To assess silver-supervised adaptation, we compare Qwen with and without a project-trained LoRA adapter (Hu et al., 2022), using the same base checkpoint and inference procedure. JobBERT-zh + CRF provides a supervised sequence-labelling baseline that combines an encoder pretrained on Chinese recruitment text with an inherited CRF head. The encoder is derived from `hfl/chinese-roberta-wwm-ext`.

To assess taxonomy-informed initialisation, we compare XLM-R-large (Conneau et al., 2020) and ESCOXLM-R (Zhang et al., 2023) with newly initialised nine-label BIO heads. The pair uses the same silver split, training settings, and checkpoint-selection criterion. Different prediction heads and training budgets make comparisons with JobBERT and Qwen descriptive.

Table 5. Evaluated models, access routes, and inference settings.

Model	Implementation	Inference settings
GPT-5.6 Terra	API	Temperature 0; 2,048 output tokens.
Claude Sonnet 4.5	API	Temperature 0; 2,048 output tokens.
DeepSeek-V4-Pro	DeepSeek API	Non-thinking: temperature 0, 2,048 tokens; thinking: 8,192 tokens, temperature unspecified.
Kimi K2.6	Moonshot API	Non-thinking; temperature 0.6; 2,048 output tokens.
Qwen (unadapted / LoRA)	Qwen2.5-14B-Instruct; local weights	Deterministic decoding; 2,048 output tokens.
JobBERT-zh + CRF	Released encoder and inherited CRF, continued on silver labels	Sequence labelling with CRF decoding.
XLM-R-large	Local encoder; new nine-label BIO head	Character-level sequence labelling.
ESCOXLM-R	Local encoder; new nine-label BIO head	Character-level sequence labelling.

Token budgets are maximum output lengths. GPT and Claude names follow service responses; upstream identities were not independently verified, and GPT request and response identifiers differ (Appendix C).

Training and Inference

Supervised checkpoints are selected by typed exact-span F1 on their respective silver development sets. JobBERT-zh + CRF continues training the released encoder and inherited CRF, using supervision informed by handbook revisions, model-generated labels, and human adjudication. Appendix C compares the supervision versions.

Qwen starts from pretrained Instruct weights with a fresh LoRA adapter for each of seeds 42, 43, and 44. Training uses a fixed system instruction, user template, and five schematic examples. Targets specify the literal span text, its type, and an index identifying which occurrence of that text in the sentence is intended (0 denotes the first occurrence). The loss is computed over the assistant's JSON output and closing turn token. Training uses BF16 precision. JobBERT and Qwen use early stopping with a patience of two epochs; Table 6 lists the other training settings.

XLM-R-large and ESCOXLM-R use AdamW and seeds 42, 43, and 44. BIO labels are assigned to each original Chinese character's first subword. Predictions from chunks of at most 128 source characters, including the remainder, cover complete sentences and are scored against the original text and offsets. Appendix C details tokenisation that preserves source positions.

Each generative request contains one sentence's identifier and text, shared instructions based on handbook v4.2.12, and five fixed examples. These archived instructions were retained for the reported evaluation; the later v4.2.14 handbook examples in Table 2 do not replace them. A common parser maps the returned literal spans and occurrence indices to Unicode-code-point offsets. Adapted and unadapted Qwen share the base checkpoint, prompts, parser, BF16 precision, deterministic decoding, and a 2,048-token output limit. Other prompted configurations share the instructions and use the inference settings in Table 5, including the output budgets used for DeepSeek with and without thinking.

Table 6. Training settings for the silver-supervised model comparisons.

Setting	JobBERT-zh + CRF	Qwen + LoRA	XLM-R / ESCOXLM-R
Learning rate	2×10^{-5}	10^{-4}	2×10^{-5}
Batch size	16	1; accumulation over 16 steps	8
Input handling	Up to 256 tokens	Up to 8,192 tokens	Chunks of up to 128 source characters
Maximum epochs	6	6	20

Qwen LoRA uses rank 16, scaling 32, and dropout 0.05 on the attention q/k/v/o projections.

Evaluation Metrics

The primary metric is typed exact-span micro-F1, which requires identical span boundaries and L/K/S/T type. Typed relaxed-span micro-F1 allows boundary variation through one-to-one greedy matching of same-type spans with intersection-over-union (IoU) of at least 0.5:

$$\mathrm{IoU}(u,v) = \frac{|u \cap v|}{|u \cup v|}, \qquad \mathrm{IoU}(u,v) \geq 0.5. \tag{1}$$

Here, u and v are the sets of Unicode-code-point positions in the predicted and reference spans; $|\cdot|$ counts positions. Under either matching rule, true positives (TP) are matched prediction–reference pairs, false positives (FP) are unmatched predictions, and false negatives (FN) are unmatched reference spans. Summing counts over sentences gives:

$$P_{\mathrm{micro}} = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FP}}, \quad R_{\mathrm{micro}} = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FN}}, \quad F_{1,\mathrm{micro}} = \frac{2\mathrm{TP}}{2\mathrm{TP}+\mathrm{FP}+\mathrm{FN}}. \tag{2}$$

All 150 reference sentences are scored. When some span candidates in a response are rejected, its accepted spans are still scored. Malformed responses and responses with no accepted spans are treated as empty predictions. Sentences with neither predicted nor reference spans contribute no span counts, and a metric with a zero denominator is set to zero. Appendix C details the parser, matching rules, and scoring code.

Boundary-only exact-span micro-F1 matches boundaries while ignoring type. Per-type scores use all reference sentences, including those without a reference span of the evaluated type. Macro-F1 averages category-level F1. Models predict four types, and the encoder comparison reports both L/K/S/T and K/S/T averages; the latter omits the two-span L category.

Main three-seed experiments are summarised by their mean and sample standard deviation. Prompted configurations and expanded-pool experiments are single runs unless stated otherwise. Paired model comparisons use 95% percentile intervals from 10,000 bootstrap resamples of the 61 source-document groups in the human reference. Each draw samples complete groups with replacement and is shared by all compared models. These intervals describe variation from resampling these reference groups for fixed model runs; seed standard deviations describe variation across training runs. Appendix C gives the full calculation.

RESULTS

Figure 5 summarises prompted inference and silver-supervised models on the common human reference set. Trained configurations are summarised over three seeds; prompted configurations are single runs. Appendix C provides detailed scores. The following analyses examine competency types, extraction behaviour, and differences between silver and human evaluation targets.

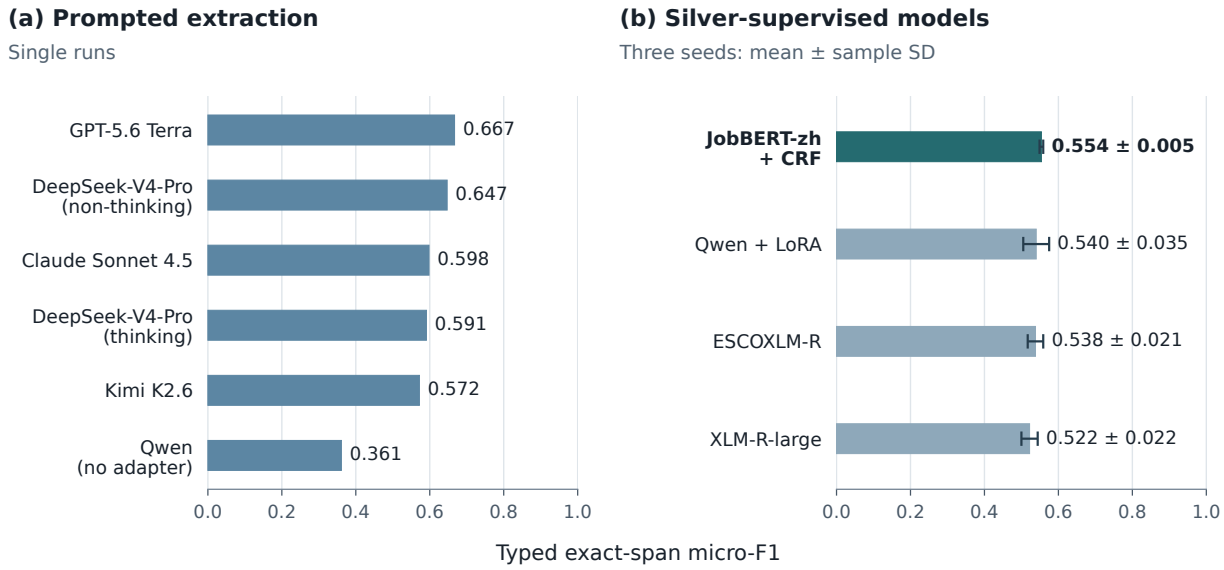


Figure 5. Typed exact-span micro-F1 on the same 150-sentence human reference set. Panel (a) shows single prompted runs; panel (b) shows means with sample standard deviations over seeds 42, 43, and 44. Training and inference settings differ across model families. Detailed scores appear in Tables 17, 18, and 8.

Prompted Inference on the Human Reference Set

GPT-5.6 Terra achieved the highest observed typed exact-span micro-F1 among six configurations, scoring 0.667 compared with 0.647 for DeepSeek without thinking (Figure 5a; Table 17). The paired F1 difference was 0.020, with a 95% confidence interval of $[-0.027, 0.061]$ obtained by bootstrapping the reference groups (Table 7). Because the interval includes zero, it does not provide clear evidence of a performance advantage for GPT.

The DeepSeek configurations differed in precision and recall. With thinking, precision was higher (0.713 versus 0.651), but recall was lower (0.505 versus 0.643), with lower exact-span F1. Their inference budgets differed. Unadapted Qwen had the lowest recall (0.281).

Silver-Supervised Adaptation

Silver supervision improved Qwen's typed exact-span micro-F1 under the fixed inference protocol. The adapted mean was 0.540 ± 0.035, compared with 0.361 without an adapter (Figure 5b; Table 18, panel A). The mean gain over the three training seeds was 0.179, with a paired 95% bootstrap interval of $[0.111, 0.242]$; the intervals for all three individual seeds also lay above zero

Table 7. Paired differences in typed exact-span micro-F1.

Comparison	ΔF_1	95% CI
GPT-5.6 Terra − DeepSeek-V4-Pro (non-thinking)	+0.020	$[-0.027, 0.061]$
Qwen LoRA (seed 42) − Qwen (no adapter)	+0.138	$[0.069, 0.200]$
Qwen LoRA (seed 43) − Qwen (no adapter)	+0.202	$[0.133, 0.269]$
Qwen LoRA (seed 44) − Qwen (no adapter)	+0.197	$[0.123, 0.268]$
Qwen LoRA (seed mean) − Qwen (no adapter)	+0.179	$[0.111, 0.242]$
ESCOXLM-R − XLM-R-large (seed means)	+0.016	$[-0.005, 0.035]$

ΔF_1: first model minus second. CI: pointwise paired source-group bootstrap confidence interval (61 groups).

(Table 7). Mean scores also increased under relaxed and boundary-only matching, showing gains across the three matching criteria.

The adapters produced fewer responses with rejected span candidates than the unadapted model. These end-to-end scores reflect both span extraction and output validity. The reproduction materials include per-seed Qwen scores and parser outcomes.

Among the silver-supervised models, the JobBERT-zh + CRF baseline reached a mean typed exact-span micro-F1 of 0.554 ± 0.005 (Figure 5b; Table 18, panel B). Its mean was above Qwen + LoRA (0.540). Both fell short of prompted GPT-5.6 Terra (0.667). These figures, however, come from evaluated configurations whose training and inference protocols differ across model families, so the gaps should be read as descriptive rather than controlled comparisons.

Encoder and Competency-Type Comparisons

ESCOXLM-R scored above XLM-R-large under their shared Chinese-supervision protocol, with mean typed exact-span micro-F1 of 0.538 and 0.522, respectively (Table 8, panel A). The paired mean difference was 0.016, with a 95% interval of $[-0.005, 0.035]$ (Table 7). The interval includes zero and does not establish an advantage for ESCO initialisation in this comparison.

Four-type macro-F1 was 0.607 for XLM-R-large and 0.622 for ESCOXLM-R; the K/S/T averages were 0.542 and 0.562. The latter summaries exclude L from the average while retaining the four-type prediction task. Each run of the three trained models in panel B recovered both reference L spans and made one false-positive prediction, yielding 0.800 ± 0.000. Zero standard deviation reflects identical counts across runs; L contains only two reference spans.

Qwen adaptation helped most for T-type competencies, where mean F1 rose from 0.380 to 0.638 (Table 8, panel B; Figure 6a). Mean F1 increased in all three K/S/T categories. The gains for S and K were 0.148 and 0.210, respectively. Within the three silver-trained configurations in panel B, Qwen had the highest observed means for S and T; ESCOXLM-R had the highest K mean. The three configurations use different prediction heads, training budgets, and inference procedures, so these are observed rankings, not controlled performance claims.

Extraction Diagnostics

The recall gains from Qwen adaptation extended across all three reference-span length bins: ≤ 4, 5–8, and ≥ 9 Unicode code points (Figure 6b). Each adapter improved exact-match recall in every bin, which contained 346, 243, and 74 reference spans, respectively.

Adaptation also reduced false-positive spans on the eight sentences with no reference spans

Table 8. Overall and per-type results on the human reference set.

A. Encoders with Chinese supervision: overall F1

Encoder	Typed exact	Typed relaxed	Boundary-only exact	Macro L/K/S/T	Macro K/S/T
XLM-R-large	0.522 ± 0.022	0.650 ± 0.021	0.571 ± 0.022	0.607 ± 0.016	0.542 ± 0.021
ESCOXLM-R	0.538 ± 0.021	0.662 ± 0.022	0.582 ± 0.024	0.622 ± 0.010	0.562 ± 0.013

B. Typed exact-span F1 by competency type

Model	S ($n = 448$)	K ($n = 125$)	T ($n = 88$)	L ($n = 2$)
Qwen, no adapter	0.377	0.302	0.380	0.000
Qwen + LoRA	0.525 ± 0.039	0.512 ± 0.063	0.638 ± 0.017	0.800 ± 0.000
XLM-R-large	0.506 ± 0.024	0.495 ± 0.037	0.627 ± 0.006	0.800 ± 0.000
ESCOXLM-R	0.518 ± 0.029	0.537 ± 0.009	0.631 ± 0.020	0.800 ± 0.000

Mean $\pm$ sample SD over three training seeds; unadapted Qwen: single run. n: reference spans.

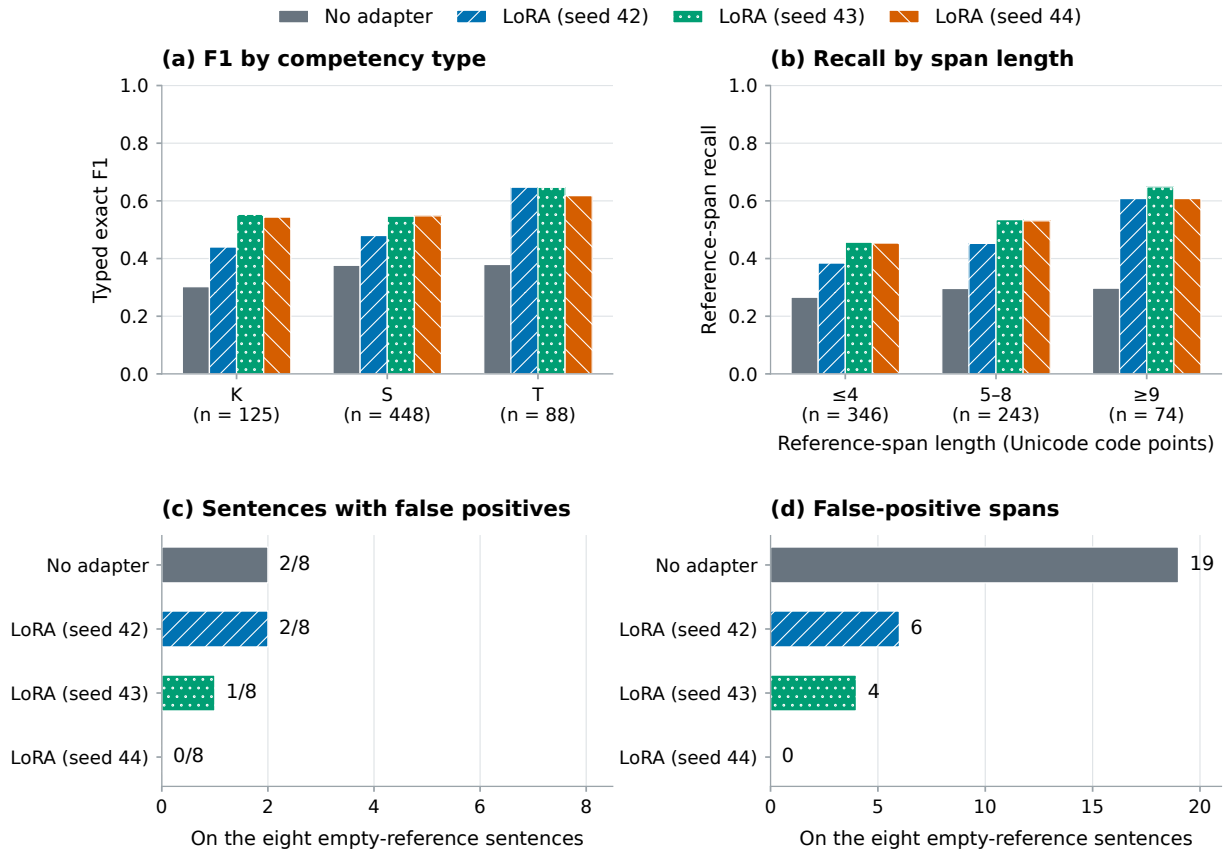


Figure 6. Qwen diagnostics on the human reference set: (a) typed exact-span F1 for K/S/T; (b) recall by reference-span length (n: reference spans); (c–d) false positives on eight sentences with no reference spans, counted by sentence and span.

(Figure 6c,d). The unadapted model produced 19 false-positive spans in two sentences, compared with 6, 4, and 0 spans for seeds 42, 43, and 44. The corresponding numbers of affected sentences were two, one, and zero. For seed 42, fewer false-positive spans occurred in the same two sentences, so the number of sentences containing false positives did not change.

The unadapted Qwen predictions illustrate how boundaries and types affect scoring. Applying relaxed matching to the same predictions raised F1 from 0.361 to 0.470 (Table 17). Under exact matching, the 181 false positives comprised 27 same-boundary/different-type pairs, 89 same-type/overlapping-boundary pairs, and 65 other unmatched predictions. These groups describe relationships to the reference annotations. Overlapping boundaries alone do not guarantee a relaxed match, which also requires the same type and an IoU of at least 0.5.

Two examples illustrate these patterns. In the first, Qwen extended an S span meaning "technical project development planning" to include surrounding action and object wording; its overlap with the reference remained below the relaxed threshold. In the second, Qwen assigned S to the word meaning "method", while the reference assigned K to the same span. Both pairs therefore failed exact and relaxed typed matching. Appendix C provides the complete Chinese sentences, predicted and reference spans, offsets, and sentence identifiers.

Evaluation with Expanded Silver Pools

The two expanded-pool Qwen configurations changed order across silver and human-reference evaluation. With seed 42, the adopted pool yielded F1 of 0.809 on its silver test set and 0.585 on the common human reference; the alternative yielded 0.759 and 0.597, respectively (Table 21). Thus, the configuration with higher agreement on its generated test labels had lower agreement on the shared human reference. The pool-specific silver tests differ in both texts and labels, so their score gap does not isolate annotation quality or the effect of the annotation channel.

The annotation-quality analysis provides a separate comparison of silver labels against blinded human annotations on common sentences. Table 22 gives the constituent batches and final counts for the expanded pools. Exploratory expanded JobBERT runs in Appendix C scored below the main baseline, with supervision, source composition, and model selection changed together.

DISCUSSION

Silver supervision improved Qwen's end-to-end extraction under matched inference conditions. All three adapters raised F1 across the K, S, and T types and recall in each of the three reference-span length bins (Table 8; Figure 6). The adapters also produced fewer responses with rejected span candidates, so the scores reflect span extraction and output validity together. These results support using silver labels to adapt a model to this annotation task. JobBERT-zh + CRF had a slightly higher mean F1 than Qwen + LoRA in the main silver-supervised comparison, although their training and inference settings differed.

Aggregate scores do not show where extraction fails. The gap between exact and relaxed F1 indicates how sensitive the results are to boundary matching. Sentence examples show whether an error is a boundary mismatch or a type disagreement (Extraction Diagnostics). For this reason we report exact, relaxed, and boundary-only scores together, and character offsets let readers find each disagreement in the source text.

A single-seed comparison of expanded silver pools (Table 21) adds a caution. The configuration that scored higher on its own silver test did not score higher on the shared human reference. This is a descriptive observation from one run, so it does not establish a ranking across training runs. The silver tests differ in both texts and annotations, whereas the human reference is a shared evaluation target.

The human reference concentrates on ambiguous cases that came up during handbook development, so the type-level results should be read with this in mind. K combines knowledge with qualifications and industry background, and the human reference contains only two L spans, so L results rest on very few cases. Reference sentences were kept out of Qwen and encoder supervision, but 59 of the 61 source-document groups identified from reference record IDs also occur in the 2,150-record silver training set (Appendix D). Scores for models trained on this set may therefore overstate performance on unseen documents. JobBERT-zh's original pretraining corpus was unavailable, so its overlap with the human reference could not be checked (Appendix D). Independent annotation of new documents would broaden the evaluation across requirement types and recruitment contexts. The annotations describe advertised requirements. Using them to assess individuals would require separate evidence of validity and fairness.

CONCLUSIONS

Chinese-SkillSpan is a benchmark for identifying competency spans and assigning their types in Chinese job advertisements. It combines source-text annotations with character-level offsets, an ESCO-informed handbook, separate human reference and silver labels, and shared scoring procedures. In a small blinded study, agreement on span boundaries alone was slightly higher than agreement that also required types, although the confidence intervals overlap.

With inference settings fixed, silver-label adaptation improved Qwen's end-to-end extraction over the unadapted model. A prompted GPT model ranked highest among the configurations evaluated, but its lead was not clear-cut. The released annotations, handbook, JobBERT-zh checkpoints, predictions, and scoring code allow others to compare new systems against the same targets.

Two extensions remain. Mapping spans to ESCO concepts, which Zhang et al. (2022b) also leave to future work, would connect the benchmark to the wider taxonomy. Allowing overlapping type layers would test whether a single flat type loses information in ambiguous cases, and it would need scoring that accepts more than one valid type for a span.

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DATA AVAILABILITY

The dataset archive, v0.1.3 (DOI: 10.5281/zenodo.22698504), contains the corpus, the Chinese v4.2.14 handbook dated 9 September 2026, the 150-sentence human reference, and 2,150/169 silver training/development records used by Qwen, XLM-R-large, and ESCOXLM-R. The later reader editions are linked in Appendix A. This version-specific DOI identifies the dataset used here; the archive series has concept DOI 10.5281/zenodo.22288337. Later historical annotation and assisted-review materials and the blinded agreement study are released separately, with pseudonymised annotator identifiers, source texts, offsets, and individual annotation layers.

The Qwen archive (DOI: 10.5281/zenodo.22851581) provides three original-silver LoRA adapters, selected expanded-silver checkpoints, predictions, configurations, and scoring tools. For the 9,540-record adopted and 9,646-record alternative expanded pools, split identifiers and selected outputs are public; complete texts and labels are available through the corresponding author for confidential peer review. Six Chinese-supervised XLM-R/ESCOXLM-R checkpoints are linked in the model index, with code, configurations, and predictions in the evaluation archive (DOI: 10.5281/zenodo.22942441). The reproduction documentation separately describes local score recalculation from saved predictions and reported results of model reruns on a server.

The `AlfredJames/jobbert-zh` repository provides the domain-adapted encoder and existing CRF used for initialisation; `AlfredJames/jobbert-zh-v6a` provides silver-trained checkpoints for seeds 42, 43, and 44 (default: seed 42). The project repository and reproduction guide provide code, procedures, and records, including supplementary external-task experiments. Table 24 summarises access and reproduction scope.

Reuse follows component-specific terms: Apache-2.0 for explicitly covered software and documentation; academic research and peer review with attribution for author-controlled annotations and research materials; source permissions for third-party recruitment texts; and accompanying notices and base-model licences for checkpoints and adapters. The data-access guide and Appendix D provide the terms, contact-pattern findings, and review limitations.

APPENDIX A. CODING GUIDE AND EXAMPLES

This appendix summarises the v4.2.14 rule series with bilingual examples. Figure 2 shows the preserved human-reference annotations. The later Chinese and English handbooks, revised 9 October 2026, provide further examples and identify cases awaiting review. Section A.4 distinguishes these reading materials from the versions associated with the reported studies.

A.1 Label definitions and notation

Table 2 defines the four labels with examples. Square brackets mark selected source substrings, followed by their type codes. Brackets, colours, and type codes are display aids rather than source characters; brackets keep boundaries identifiable in greyscale. The examples illustrate annotation decisions and are not an additional evaluation sample.

Source key. E: ESCO's skills hierarchy and transversal concepts (European Commission, 2024); Q: EQF definitions (Council of the European Union, 2017); Z: related annotation guidance in SkillSpan, Appendix B (Zhang et al., 2022b); I: unitisation and categorisation in agreement analysis (Artstein and Poesio, 2008); P: our V4.2.14 operational decision. Combined marks indicate adaptation, not verbatim rules; Chinese examples and exact boundaries are project illustrations.

A.2 General annotation decisions

G1–G15 are quick-reference identifiers. Their source sections and amendment IDs (R08–R23) are mapped in the release; they do not replace the handbook's rule numbering.

Table 9. Annotation principles for competency mentions and span boundaries.

Rule	Annotation principle	Example and English gloss
G1	The annotation scheme distinguishes L, K, S, and T. Collapsed categories are used only in secondary evaluation. [E,P]	[英语六级] [L] *College English Test Band 6* [计算机专业] [K] *Computer science major* [数据分析] [S] *Data analysis* [责任心] [T] *Sense of responsibility*
G2	Final annotations contain non-overlapping spans. A sentence may contain several independent mentions, with each character assigned to at most one span. [P]	熟悉使用 [Word] [S]、[Excel] [S]、[PPT] [S] 等办公软件 *Familiar with using Word, Excel, PowerPoint, and other office software* No additional covering list span.
G3	Each mention is a contiguous source substring that preserves spelling and internal spaces. Any normalisation is stored separately. [Z,P]	熟悉 [Jekins] [S] *Familiar with Jekins* Keep the source spelling, not `Jenkins`.
G4	Annotators select the shortest span that expresses a complete requirement. Words and identifiers remain intact; semantic completeness takes priority over length, with no fixed length limit. [Z,P]	[支持服务] [S] *Support services* The truncated 支持服 is incomplete.
G5	Generic proficiency triggers usually remain outside the span, while heads needed to express the requirement remain inside. Excluded triggers can still inform type assignment. [Z,P]	会使用 [Excel] [S] *Able to use Excel* [会聊天] [T] *Can hold a conversation* 有 [服务意识] [T] *Has a service-oriented mindset*
G6	Coordinated items are separated when each remains meaningful in context. Shared qualifiers may stay outside the spans, and implied words are not added. [Z,P]	熟悉使用 [Word] [S]、[Excel] [S]、[PPT] [S] 等办公软件 *Familiar with using Word, Excel, PowerPoint, and other office software* Shared-experience case: see A.3.
G7	A necessary action, object, or shared head remains within the span when splitting would change its meaning. Span length alone does not determine type. [Z,P]	[优化曝光与转化率] [S] *Optimise exposure and the conversion rate* Isolated metric names would lose the optimisation activity.

Table 10. Annotation principles for scope, experience, type, and unresolved records.

Rule	Annotation principle	Example and English gloss
G8	Benefits, company promotion, administrative instructions, experience durations, and pure outcomes are outside competency scope on their own. Mixed clauses are assessed separately. [Z,P]	五险一金，带薪年假 → []; *Five types of social insurance and a housing fund; paid annual leave* [分析推广效果] [S]，保证流量/用户增长 *Analyse the effectiveness of promotional activities; ensure traffic/user growth*
G9	Experience wording is omitted only when the remaining activity preserves the requirement. Necessary practical-experience wording is retained. Explicit industry background follows the handbook's broad-K convention. Experience wording is never added to the source. [P]	[机器人竞赛经验] [S] *Experience in robotics competitions* Retain the suffix if a bare competition name changes the requirement; do not label generic 经验 (experience) alone.
G10	Educational qualifications include their threshold wording. Language examinations and certificates receive L; non-language qualifications receive K. [Z,P]	[本科及以上学历] [K] *Bachelor's degree or higher* [大学英语 6 级] [L] *College English Test Band 6* 了解 [ISO 27001] [K] *Familiar with ISO 27001* This denotes familiarity with the standard rather than personal certification.
G11	Tool and method mentions are typed from their predicate context. Occupational use supports S; principles or theory support a complete K phrase. Familiarity alone does not determine type. [Q,P]	使用 [SQL] [S] *Use SQL* 了解 [SQL 原理] [K] *Understand the principles of SQL*
G12	Occupational functions are distinguished from general cognition, communication, and coordination. A duty clause may contain T; usefulness across industries alone does not establish T. [E,P]	[编码能力] [S] *Coding skills* [沟通能力] [T] *Communication skills* [协调外部资源] [T] *Coordinate external resources*
G13	Scope, boundaries, and type are resolved separately, without a fixed priority among labels. If only the type remains uncertain, a mention may receive provisional S with alternatives recorded. Such unresolved records are excluded from final training targets. [I,P]	Heading alone: 能力提升 *Developing capabilities* The heading alone leaves scope unresolved; it does not establish an empty annotation.
G14	Offsets count original Unicode code points, beginning at zero and excluding the end position. The indexed substring must reproduce the mention exactly; segmentation is a derived view. It does not replace the fixed human-reference offsets. [P]	`text[start:end] == span` Keep the original text, internal spaces, and offsets consistent.

Table 10 (continued)

Rule	Annotation principle	Example and English gloss
G15	Alternatives and overlaps are recorded separately from the final flat layer. An unresolved record is excluded in full from training, with confirmed empty, missing, and unresolved annotations distinguished. [P]	五险一金 *Five types of social insurance and a housing fund* → confirmed `[]`; 能力提升 *Developing capabilities* Without context → unresolved. Known partial spans do not make an unresolved record training-ready.

Interpreting coordinated capabilities. G6–G7 test whether each selected phrase expresses a requirement in context. Items need not repeat an implied capability noun. For example, 协调 (*coordination*) can express a capability independently, whereas 曝光 (*exposure*) alone does not express the optimisation action in 优化曝光与转化率 (*optimise exposure and the conversion rate*). The figure preserves the human-reference boundaries used for scoring; the examples below illustrate the rules summarised here. Their respective handbook versions are identified in Section A.4.

A.3 Shared experience: splitting without changing the requirement

The V4.2.14 amendment (R23) clarifies a shared-experience construction. [P] Each item remains identifiable in context; exclude the shared experience qualifier and do not add a covering span. The words for internet-company operations denote an occupational activity in this sentence; a bare requirement for experience in the internet industry would instead use the project K proxy rule. The presence of an industry name alone does not determine type.

Chinese. 4、 三 年 以 上 大 中 型 [互联网公司经营] [S] 或 [BI] [S] 相 关 工 作 经 验，[互联网公司广告运营分析] [S] 优先；

English. At least three years of relevant work experience in [internet company operations] [S] or [BI] [S] at medium-sized or large companies; experience in [advertising operations analysis at internet companies] [S] preferred.

Highlights align the three S spans across languages. Chinese offsets: $[9,16)$, $[17,19)$, and $[26,37)$; the original item number is retained. BI is S in this context, not by a universal dictionary rule. In contrast, G7 retains the shared action in 优化曝光与转化率 (*optimise exposure and the conversion rate*); splitting must not leave bare outcomes. The initial reviewed LLM annotations retain the earlier long span.

A.4 Handbook Versions Used in This Study

Table 11 links annotation and evaluation stages to their documented handbook versions. Label metadata and operating instructions provide different evidence of version use; individual silver training and development records have no handbook-version field.

The bulk annotation prompt and handbook index provide the study materials and dated revisions. Several revisions share the v4.2.14 number and are distinguished by edition date and file. Later reader editions clarify rules for new annotation without replacing preserved study labels.

Table 11. Handbook versions associated with the reported annotation and evaluation stages.

Stage	Handbook record	Interpretation
Human reference	v4.2.10; v4.2.11	All 150 records carry v4.2.10 metadata; reference-set documentation identifies v4.2.11 as an operating handbook.
Initial silver review	v4.2.10 rev1; v4.2.11	The initial 100-sentence prompt uses v4.2.10 rev1; later coding and adjudication records identify v4.2.11.
Bulk silver generation	v4.2.12 rev2	Basis of the archived prompt for the remaining 2,351 original silver records.
Model evaluation	v4.2.12-based instructions	Archived shared generative instructions; later handbook examples do not replace them.
Blinded-study preparation	v4.2.14 series	Training materials are preserved. The exact handbook file used for formal coding has not been established.

The shared-experience example in Section A.3 illustrates a later boundary clarification; its original training annotation is retained. Figure 2 reproduces the human-reference labels, whereas Table 2 and Tables 9–10 illustrate the v4.2.14 rules summarised here. Batch records, version history, and the corresponding example are documented in the annotation documentation.

Archived predictions can be rescored with their reference and the fixed scorer. New model outputs or annotations may vary with handbook text, model versions, or service settings.

Figure 7 illustrates further type and boundary decisions. Panel (a) assigns T to conversational ability, service awareness, interpersonal communication, and coordination. Panel (b) retains the historical S spans for presales consulting, project management, and presales work on large projects. The English translation preserves the five-year minimum, quantity, and experience qualifiers as unhighlighted context outside these spans.

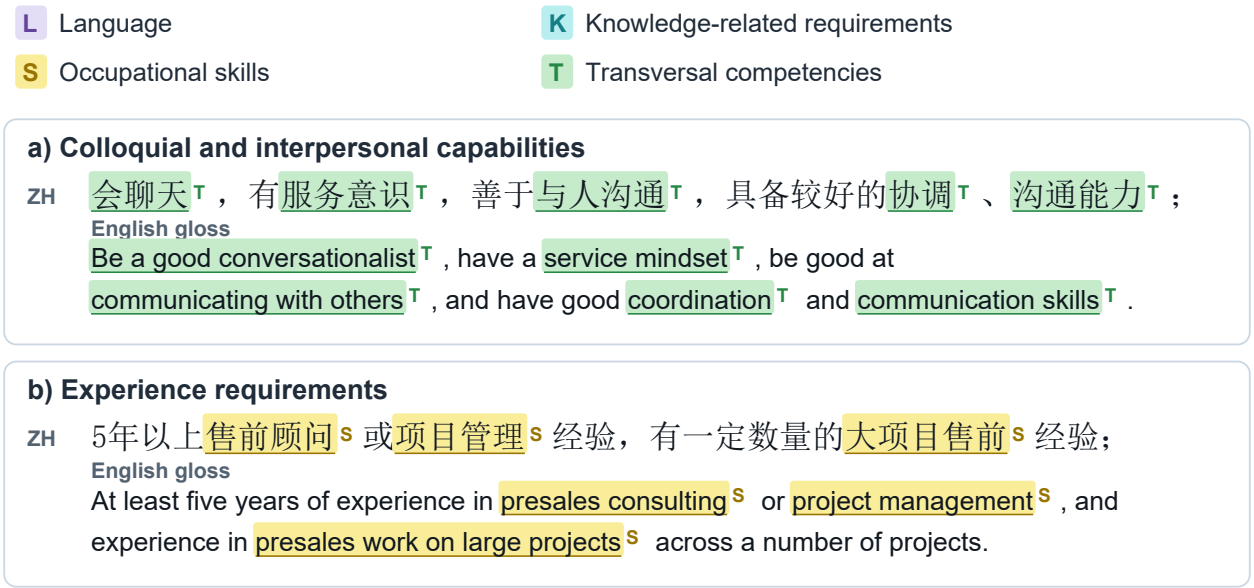


Figure 7. Additional human-reference examples. Chinese highlights retain the frozen reference spans (v4.2.10 metadata); colours and letters align their English glosses. Source IDs and offsets are in the example metadata.

APPENDIX B. ANNOTATION-QUALITY EVIDENCE

We describe reference construction, assisted review, and the separate blinded study (Table 4).

Human Annotation and Review Coverage

The 150-sentence human reference, initial 100-sentence silver review, and separate 100-sentence post-generation review cover 350 unique sentences. The early 15-sentence assisted-review subset is included in the post-generation cohort. The additional review contributes 150 sentences: 60 cloud, 45 public-sector, and 45 listed-company sentences. Identifier and normalised-text checks excluded the original 350 sentences from this additional sample, yielding 500 sentences with human annotation or review before the blinded study. The 50-sentence blinded study has no normalised-text overlap with these cohorts, bringing coverage to 550 distinct sentences. Each sentence is counted once across annotation layers. Model-evaluation labels come from the 150-sentence human reference; the remaining cohorts serve annotation-quality assessment.

Reference Construction and Initial Agreement

The human reference combines 50 calibration sentences sampled from the 980-record model-disagreement queue and a separate 100-sentence challenge sample. The calibration sentences were annotated separately and adjudicated; the challenge sample was collaboratively annotated and reviewed. The authors recall machine suggestions being available during calibration, whereas an earlier protocol specifies raw-text-only coding. The available records do not resolve this discrepancy, so this stage is not treated as a blinded agreement study. All 50 calibration sentence IDs are absent from the silver training and development sets.

The finalised reference contains 415 challenge-sample spans (290 S, 83 K, 41 T, one L) and 248 calibration-sample spans (158 S, 42 K, 47 T, one L). Seven challenge sentences and one calibration sentence have no annotated span. The 150 reference sentences correspond to 61 source-document prefixes in the original corpus. These reference labels are used for model evaluation and remain separate from the individual coding and assisted-review layers.

The two available calibration annotation layers contain 211 and 230 spans, with 98 exact typed matches and F1 of 0.444. Character-level κ is 0.567 across all 2,571 source code points, including spaces and punctuation. These retrospective estimates use available exports whose correspondence to the original frozen files has not been verified. They describe initial agreement, rather than an independent assessment of the adjudicated reference. Comparisons with that reference involve the same annotators and reflect adjudication changes; the detailed scores and supporting records are provided in the annotation documentation.

Character-level Cohen's κ adjusts observed agreement for agreement expected by chance from the two coders' marginal label frequencies across the source character positions:

$$\kappa = \frac{p_o - p_e}{1 - p_e}, \qquad p_e = \sum_{c \in \mathscr{C}} p_A(c)\, p_B(c). \tag{3}$$

Here, p_o is the fraction of source character positions with identical labels, and $p_A(c)$ and $p_B(c)$ are the two coders' label proportions. The label set $\mathscr{C}$ comprises L/K/S/T and O. For the four-label scheme plus O, the available exports yield $p_o = 0.771$ and $p_e = 0.471$, giving $\kappa = 0.567$. These coefficients use character positions rather than the spans counted in Eq. (2). The 2,571 positions are nested within the 50 sampled sentences.

When both annotations of a sentence are empty, the pair contributes no span true positives. Span matching and character-level agreement use different units; optional category projections are described in the reproduction materials.

Post-Generation Assisted Review

The 100-sentence post-generation review sample was stratified by source, empty/nonempty annotation, and text length. The sampling frame contained 2,179 eligible records after excluding 100 initially reviewed records, 100 reserved for an earlier pilot, and 72 pending records from the 2,451-record original silver pool. Two small strata totalling 11 records were unsampled.

Comparing machine-prefilled and human-reviewed annotations yields 140 exact typed matches among 144 machine and 148 reviewed spans: precision is 0.972, recall 0.946, and F1 0.959. Ninety-five sentences have identical span sets. These estimates apply to this stratified sample.

The 15-sentence assisted-review subset is nested within the post-generation Silver review cohort. Both annotators had access to machine coding suggestions. Their human layers contain the same 27 spans, yielding exact span F1 and character kappa of 1.000. The shared suggestions may contribute to this agreement between assisted reviewers.

Supervision Counts and Length Summaries

Figure 4 compares the supervision and human-reference distributions. The median input length is 33 Unicode code points in the silver training and development sets used by Qwen, XLM-R-large, and ESCOXLM-R, and 41.5 in the human reference. Median span lengths are six and four code points, respectively. The training and development sets contain 3,646 and 345 spans; 869 of 2,150 training records (40.4%) and 82 of 169 development records (48.5%) have no annotated span. Table 12 reports counts, including repeated records, for the adopted expanded pool used in the supplementary Qwen experiments.

Table 12. Competency-span counts in the adopted expanded silver pool and its 8,842/348/350 training/validation/test splits for the supplementary Qwen experiments.

Split	Records	Empty annotations *n* (%)	L	S	K	T	Total spans
Training	8,842	2,321 (26.2%)	149	10,653	4,887	3,851	19,540
Validation	348	91 (26.1%)	2	403	184	152	741
Test	350	80 (22.9%)	4	472	194	150	820
Full pool	9,540	2,492 (26.1%)	155	11,528	5,265	4,153	21,101

Percentages use the number of records in each split.

Additional Review of Newly Generated Silver Labels

In the additional assisted review, two reviewers separately annotated the same 150 sentences with Codex and Cursor suggestions visible (Table 19). Figure 8 shows their shared input-length distribution and separate span-length distributions. Median input length is 33.5 Unicode code points; both reviewers' median span length is six, with maxima of 41 for A and 42 for B. Lengths count spaces and punctuation; the input distribution includes sentences with empty annotations.

Table 13 gives both reviewers' type counts. Table 14 compares pre-adjudication span sets. Its 95% percentile intervals use 10,000 sentence resamples within source strata, with fixed quotas.

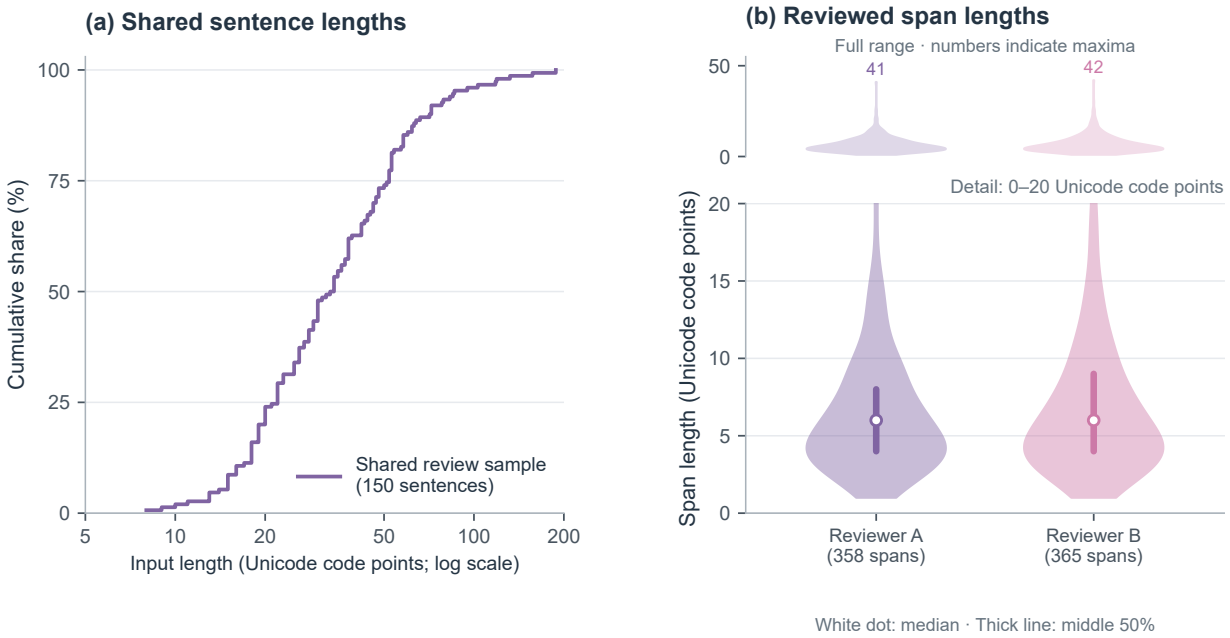


Figure 8. Length distributions in the additional assisted-review sample. Panel (a) uses the same 150 sentences for both reviewers; panel (b) shows their separate annotation layers, with 358 spans for A and 365 for B. Densities use all observed spans with equal maximum widths. The overview shows the full range of span lengths, while the detail view focuses on 0–20 Unicode code points. Dots mark medians, and thick lines mark the middle 50%. Supervision and human-reference distributions appear in Figure 4; reviewer-specific type counts appear in Table 13.

Table 13. Competency composition in the additional review sample: span counts (within-reviewer percentages).

Reviewer	L	S	K	T	Total spans
A	3 (0.8%)	217 (60.6%)	77 (21.5%)	61 (17.0%)	358
B	3 (0.8%)	218 (59.7%)	81 (22.2%)	63 (17.3%)	365

Table 14. Pre-adjudication typed exact-span agreement in the additional review sample.

Comparison	Exact F1	95% CI	Sentences with identical span sets
Reviewer A–Reviewer B	0.816	[0.769, 0.860]	101/150
Codex–Cursor suggestions	0.962	[0.937, 0.983]	138/150
Reviewer A–Codex suggestions	0.823	[0.776, 0.866]	100/150
Reviewer B–Codex suggestions	0.850	[0.803, 0.892]	108/150

F1 is micro-averaged; CIs are 95% percentile intervals from a source-stratified sentence bootstrap.

Agreement uses both reviewers' complete annotation layers, including the 49 sentences with differing span sets. Character-level calculations assign O/L/K/S/T to the original Unicode-code-point positions, including spaces and punctuation. Two positions have conflicting types within one reviewer's layer; excluding them from both character sequences leaves $N = 5{,}990$ positions. All original spans remain in span-level F1. The accompanying agreement data identify the affected record. Later adjudication and incorporation into training targets have not been verified. Agreement over the retained character positions is $p_o = 0.934$, with $p_e = 0.439$ and $\kappa = 0.883$.

For nominal Krippendorff's α, let m_c count assignments to category c across both reviewers, with $\sum_c m_c = 2N$. The observed and expected disagreements are

$$\alpha = 1 - \frac{D_o}{D_e}, \qquad D_o = 1 - p_o, \qquad D_e = 1 - \frac{\sum_c m_c(m_c - 1)}{2N(2N-1)}. \tag{4}$$

Here $D_o = 0.066$ and $D_e = 0.560$, giving $\alpha = 0.883$. Character coefficients include the many O positions, whereas span F1 assesses complete spans (Artstein and Poesio, 2008). The machine–machine comparison measures concordance between the suggestion layers.

Blinded Coding by Three Annotators

The three annotators were a doctoral student in human resource management and two working professionals pursuing master's degrees in management. The two working professionals had sustained annotation experience and three to five years of professional experience. All three contributed to handbook development; two professors advised on coding and handbook preparation.

The formal sample comprises eight listed-company, 16 public-institution, and 26 Tianchi sentences from the silver annotation candidates, with one sentence per advertisement identifier. The team reports random selection, although the candidate frame and sampling log are unavailable; the released annotations identify the sampled sentences and sources. Preparation used five familiarisation and 15 practice sentences, all excluded from the agreement analysis. The available records do not establish whether all annotators followed the same sequence.

The annotators coded independently and could consult the handbook and revise their own labels before submission, with machine suggestions and other annotators' labels hidden throughout. The analysis uses the final independent annotations and includes all 50 sentences and all annotator pairs. Mean pairwise typed exact-span F1 is 0.818; the annotations were not adjudicated into a consensus reference. Intermediate annotation exports yielded 0.663 and remain available alongside the final annotations in the reproduction notes. They are retained for provenance, not treated as a separate study stage or a comparable before-and-after experiment. Appendix A.4 distinguishes the handbook records from later reader editions. The sample overlaps with expanded-model training data (Appendix D); historical model-reference labels remain unchanged.

Annotators A, B, and C produced 104, 123, and 113 spans, respectively. Exact typed matches number 90 for A–B, 92 for A–C, and 96 for B–C. O covers 63.2–64.1% of character positions.

For typed exact-span F1, a match requires identical boundaries and types, and the two annotators are treated symmetrically (Hripcsak and Rothschild, 2005). Boundary-only exact-span F1 ignores type. Character-level agreement uses pairwise Cohen's κ and three-annotator nominal Krippendorff's α on L/K/S/T/O assignments to all 2,111 source Unicode code points; O denotes positions outside annotated spans. Character-level coefficients summarise agreement at each position, while span-level F1 requires complete span matches (Artstein and Poesio, 2008).

The 95% percentile confidence intervals use 10,000 sentence-bootstrap samples drawn within source strata, with all three annotation layers retained together for each sampled sentence. They summarise variation within the sampled strata for these three annotators. Table 15 gives the pairwise comparisons underlying Table 4.

Table 15. Pairwise agreement on the 50-sentence blinded sample.

Pair	Typed exact F1 [95% CI]	Boundary-only exact F1	Cohen's κ	Identical span sets (sentences)
A–B	0.793 [0.694, 0.884]	0.828	0.880	36/50
A–C	0.848 [0.752, 0.926]	0.876	0.930	39/50
B–C	0.814 [0.721, 0.899]	0.847	0.909	37/50

F1 compares complete spans; identical span sets include empty sets. Cohen's κ uses L/K/S/T/O labels at source character positions. CIs are 95% percentile confidence intervals.

Mean pairwise typed exact-span F1 was 0.818 (95% CI [0.732, 0.894]), and boundary-only exact-span F1 was 0.850 (95% CI [0.766, 0.923]). Pairwise F1 ranges were 0.762–0.805 for K, 0.778–0.869 for S, and 0.865–1.000 for T. Character-level α was 0.907 (95% CI [0.851, 0.954]).

All three annotators produced identical complete typed-span sets for 34 sentences, including 11 with no annotated spans. They also identified the same two L spans, both in one sentence, providing limited evidence about agreement on this category. Individual annotation exports and diagnostics are available in the reproduction materials.

Silver Labels against Blinded Annotations

We compared the final expanded-pool labels with each of the three blinded annotation layers on the same 49 retained sentences (Table 16). One of the 50 study sentences remained unresolved at the candidate stage and was absent from both final pools; it was excluded from this comparison rather than scored as an empty annotation. Each per-annotator score pools exact boundary-and-type matches across the 49 sentences, and the mean averages the three micro-F1 scores.

Table 16. Agreement between final expanded Silver labels and individual blinded annotations.

Silver pool	Sentences	Annotator A	Annotator B	Annotator C	Mean
Adopted expanded pool	49	0.724	0.681	0.721	0.709
Alternative expanded pool	49	0.723	0.716	0.721	0.720

Values are typed exact-span micro-F1; the mean is taken across annotators A–C.

This post hoc analysis compares historical silver labels with human annotations made without access to machine suggestions. It complements the assisted reviews, but its findings apply to the selected sample and participating annotators and do not estimate accuracy across the full silver pool. Counts, input hashes, and code are provided in the reproduction materials. The original 150-sentence model scores are unchanged.

APPENDIX C. SUPPLEMENTARY EVALUATION

This appendix covers the shared protocol, uncertainty, errors, the JobBERT supervision comparison, and exploratory expanded-pool results. The supporting records give per-run results, scorer details, and earlier protocol comparisons across output formats and demonstration settings.

Detailed Human-Reference Model Results

Tables 17, 18, and 8 complement Figure 5 with detailed model and type-specific scores.

Table 17. Prompted extraction on the human reference set (single runs).

Model configuration	P	R	Exact F1	Relaxed F1
GPT-5.6 Terra	0.676	0.658	0.667	0.766
DeepSeek-V4-Pro (non-thinking)	0.651	0.643	0.647	0.761
Claude Sonnet 4.5	0.598	0.597	0.598	0.709
DeepSeek-V4-Pro (thinking)	0.713	0.505	0.591	0.671
Kimi K2.6	0.581	0.563	0.572	0.714
Qwen	0.507	0.281	0.361	0.470

P/R: typed exact-span precision/recall; F1: typed span micro-F1. Qwen uses no adapter.

Table 18. Human-reference results for Qwen adaptation and silver-trained JobBERT.

A. Qwen: fixed inference protocol

Condition	Train / dev records	Typed exact	Typed relaxed	Boundary-only exact
No adapter	—	0.361	0.470	0.414
Qwen + LoRA	2,150 / 169	0.540 ± 0.035	0.662 ± 0.033	0.571 ± 0.038

B. Silver-trained JobBERT baseline

Model	Train / dev records	Typed exact
JobBERT-zh + CRF	2,156 / 169	0.554 ± 0.005

Micro-F1: mean $\pm$ sample SD (seeds 42, 43, 44); unadapted Qwen: single run. Boundary-only F1 ignores type.

Shared-Guideline Evaluation against the Human Reference

The shared protocol evaluates six configurations on all 150 reference sentences, using V4.2.12-based general instructions and five schematic examples. The system prompt, examples, and checksum manifest are included in the Qwen reproduction archive. For this check, we applied Unicode Normalization Form KC (NFKC), which treats compatible forms such as full-width and half-width letters alike, and removed whitespace. None of the five complete examples matched a reference sentence or appeared in full within one. This comparison checks text overlap; it does not establish that earlier protocol choices were made independently of the reference set.

Each response identifies the sentence and gives each span as its literal text, zero-based occurrence index, and L/K/S/T type. The parser converts these spans to Unicode offsets for exact and relaxed scoring against the unchanged reference texts.

Qwen adaptation uses the fixed settings described under Training and Inference. The API configurations share task instructions but use the output budgets listed in Table 5.

For the XLM-R-large and ESCOXLM-R comparison, tokeniser placeholders retain whitespace and other source positions that would otherwise be discarded. Labels are attached to each original character's first subword, and predictions cover all chunks, including the final remainder. Scoring uses the original text and Unicode-code-point offsets.

Span parsing and matching. After mapping spans to source offsets, the parser orders candidates by start offset and original output order. It keeps the first compatible candidate and rejects duplicates, conflicting types at the same boundary, and overlaps. Partial responses retain valid spans; fully rejected or malformed responses yield empty predictions. All inputs are scored.

Exact matching deduplicates identical span tuples within each sentence. Relaxed matching traverses reference spans in source order and selects the unused same-type prediction with highest IoU, taking the first prediction on a tie. This greedy matching uses an IoU threshold of 0.5. A sentence contributes no span counts when both its prediction and reference contain no spans. The released scorer, evaluation script, and output audit provide the implementation and response counts. In the DeepSeek thinking run, 18 requests exhausted the output budget before a final answer. These responses count as empty predictions; all 150 reference sentences remain. Scores thus include failure to complete within that budget.

Run variability and paired bootstrap intervals. For configurations repeated across three training seeds, we report the mean and sample standard deviation.

To account for dependence among sentences from the same source document, we grouped the 150 reference sentences into 61 document groups. Each of 10,000 paired bootstrap draws samples 61 complete groups with replacement and includes all their sentences. The same draw is used for all compared models and each fixed training seed; micro-F1 is recomputed before seed means and paired differences are calculated. Table 7 reports percentile intervals. The intervals account for within-document dependence and are conditional on the reference set and fitted models. They are unadjusted for multiple comparisons. Training-seed SD separately measures variation across training runs. As a sensitivity check, widening the bootstrap intervals for the six displayed comparisons using Bonferroni tail probabilities does not change which intervals include zero; detailed intervals and code are provided in the supplementary calculations.

Mismatch examples. The two examples in the Results section use Qwen predictions without an adapter under the shared prompt. Sentence IDs identify the source sentences in the released reference set. The first example (sentence ID: `1801-s0004`) is （4）负责参与公司无人机飞控系统技术项目开发策划、总体方案论证、技术设计 (responsible for participating in technical project development planning, overall design assessment, and technical design for the company's unmanned aerial vehicle flight-control system). The predicted S span is 参与公司无人机飞控系统技术项目开发策划 at $[5,24)$; the overlapping reference S span is 技术项目开发策划 at $[16,24)$. Their IoU is $8/19 < 0.5$, so the pair fails both exact and relaxed typed matching.

The second example (sentence ID: `1802-s0005`) is 6、培训其他方法、工具和技术支持多个应用。(approximate gloss: training in other methods, tools, and techniques to support multiple applications). Both layers select 方法 (method) at $[6, 8)$, with predicted type S and reference type K. Offsets are zero-based, half-open Unicode-code-point intervals. The second response also contains two span candidates rejected because their occurrence indices could not be resolved in the source sentence. Both examples use the existing reference annotations.

Per-seed Qwen results. Per-seed and calibration/challenge scores appear in the per-seed record, with full-precision type-level results in the model-diagnostic records. Boundary-only exact F1 (Table 18) uses the same accepted spans without types or repeated coordinates per sentence.

Model initialisation and annotation roles. Table 5 summarises the evaluated models and their configurations. Exact API identifiers, including the difference between the GPT request and response identifiers, are documented in the model provenance record. Service-reported model identities were not independently verified against vendor APIs.

Table 19 distinguishes silver generation from assisted review. Codex and Cursor are interfaces. Codex annotation and GPT API evaluation use different configurations. Shared guidelines and visible machine suggestions may align annotation choices across configurations.

Table 19. Models and interfaces used for annotation.

Annotation task	Model	Interface
Original silver generation	GPT-6 Astra	Codex
Expansion comparison	GPT-6 Astra	Codex / API
Additional assisted review	GPT-6 Astra; Grok 4.6	Codex / Cursor
Other annotation support	Qwen; DeepSeek; Kimi	—

The expansion comparison replaces one 2,500-candidate batch while retaining earlier components. Assisted reviewers could see model suggestions; Grok used high reasoning effort. Versions of the other supporting models were incompletely recorded.

Qwen Prediction Diagnostics

Figure 6 analyses accepted predictions on the 150 reference sentences. Typed exact matching is completed on each full sentence before aggregation by reference-span length. Bins (≤ 4, 5–8, ≥ 9 Unicode code points) are fixed in advance; they contain 346/243/74 reference spans in 102/110/56 sentences, respectively. A sentence may contribute reference spans to multiple bins. Type-specific unmatched predictions count as FP even in empty-reference sentences. Results for each training seed are evaluated separately on the same reference sentences.

On the eight empty-reference sentences, accepted false-positive spans decrease from 19 without an adapter to 6, 4, and 0 for seeds 42, 43, and 44; affected sentences number 2, 2, 1, and 0.

L has two reference spans: the no-adapter run recovers 0/2 and each LoRA run 2/2, with one extra L prediction in every run. Counts of true positives, false positives, false negatives, valid empty responses, and rejected spans are available in the diagnostic records.

JobBERT under Different Label Versions

Supervision and model selection. The two JobBERT conditions start from the same released encoder and inherited CRF. Earlier supervision (archived as `B1`) uses historical extraction labels,

whereas revised supervision (archived as B2) incorporates updated guidelines, LLM annotations, and adjudication. The conditions use the same training and development texts, with label changes in 1,440 training records and 107 development records. Checkpoints are selected by typed exact-span F1 against the corresponding development labels. The comparison therefore changes both training supervision and checkpoint-selection targets. Both conditions use the JobBERT training settings in Table 6. The revised-label condition is the Silver-trained JobBERT-zh + CRF baseline reported in the main results. Qwen, XLM-R-large, and ESCOXLM-R use 2,150 training records and the same 169 development records with revised labels, after six training records were removed because their texts matched development sentences after NFC normalisation (Experimental Setup). The JobBERT label-version comparison retains its original 2,156/169 inputs; neither split is the full 2,451-record silver candidate pool.

Table 20. JobBERT performance with earlier and revised labels on matched texts.

Label version	Train / dev records	Typed exact
Earlier labels	2,156 / 169	0.142 ± 0.014
Revised labels	2,156 / 169	0.554 ± 0.005

Typed exact-span micro-F1 on the human reference set; mean $\pm$ sample SD over seeds 42, 43, and 44. Both training labels and checkpoint-selection labels differ between conditions.

The revised-label setting exceeds the earlier setting in human-reference F1 (Table 20).

Exploratory Expanded-Silver Experiments

The expanded Qwen experiments use 8,842/348/350 training/validation/test records from the adopted silver pool and 8,943/351/352 from the alternative pool. Both add annotations to earlier supervision and differ in annotation channel and included records. Shared records retain their split assignments. Checkpoints are selected by typed exact-span F1 on silver validation labels. The silver tests use separate sentence splits whose source-document groups overlap with training. Fitting and model selection exclude human-reference sentences (Appendix D).

The expanded Qwen runs use seed 42, the shared prompt and parser, the output format based on span text and occurrence indices, and the training settings in Table 6. The loss is computed only over the assistant's output. Training uses early stopping with a patience of two epochs. Both Qwen checkpoints are selected at epoch one by exact F1 on the silver validation set; decoding is deterministic with a 2,048-token output cap. Full settings are available in the experiment archive. The expanded JobBERT runs retain the domain-adapted encoder and inherited CRF. The adopted-pool JobBERT run uses 8,842 training and 698 development records. Qwen instead uses 348 validation and 350 test records with the same 8,842-record training set.

Table 21. Expanded-pool Qwen results (seed 42).

Pool variant	Silver validation agreement	Silver test agreement	Human reference exact	Human reference relaxed
Alternative	0.783	0.759	0.597	0.713
Adopted	0.833	0.809	0.585	0.703

Typed span micro-F1 (exact unless marked relaxed). Training/validation/test sizes: alternative, 8,943/351/352; adopted, 8,842/348/350.

The annotation-channel comparison replaces a 2,500-candidate batch (Table 22). The adopted-pool configuration has higher agreement with its silver test labels but a lower observed exact F1 on the human reference than the alternative-pool configuration. The replaced batch contains 2,353 accepted Codex annotations and 2,459 accepted annotations obtained through the API, respectively. The score difference thus reflects changes in both labels and included records.

On the human reference set, expanded JobBERT obtains exact F1 of 0.242 with the alternative pool and 0.232 with the adopted pool (seed 42); the latter has relaxed F1 of 0.485. A separate configuration trained on 7,117 newly annotated records gives 0.237 ± 0.002 across three seeds. All are below the original Silver-trained JobBERT-zh + CRF result. Alongside training size, these experiments change the annotation protocol, source composition, and development set.

For comparison, Qwen trained on the original 2,150-record Silver set scores 0.500 at seed 42 and averages 0.540 across three seeds. Expanded-pool Qwen therefore scores higher in these runs, whereas expanded JobBERT scores lower. Because supervision and checkpoint-selection targets change together, the results support an exploratory comparison of complete configurations. Further controlled experiments are needed to explain the JobBERT decline, including possible effects of label versions, input windowing, and decoding. The supplementary materials report exploratory single-seed supervision-subset experiments with different checkpoint-selection rules.

APPENDIX D. DATA LINEAGE, ACCESS, AND REPRODUCTION

Data Availability and Table 24 summarise the released materials and access conditions. The supplementary audit package provides supporting calculations and archived experiment records.

The terms `Gold150`, `QA150`, and `Silver-plus` denote the human reference, additional review sample, and original silver candidate pool, respectively. The main silver-supervised baselines use model-specific training/development subsets of this pool with revised labels (Appendix C).

Relationship to the Earlier Preprint

The present article substantially revises the resource definitions, annotation procedures, and evaluations in the earlier preprint (Li et al., 2026). The preprint title uses "ESCO-aligned"; the current task assigns ESCO-informed types without ESCO concept identifiers. The sample list and original annotations for the earlier 100-sample agreement report have not been recovered, so those results are not used to establish reliability for the present reference. The available experiment records do not substantiate the earlier descriptions of linking spans to concepts or evaluating on industries and time periods excluded from training. The current article therefore makes neither claim. Model results use the reference labels, supervision sets, and scoring procedures documented for each experiment.

Source Categories and Data Preparation

The source corpus was originally partitioned into 17,460 training, 2,143 development, and 3,237 test sentences, with 1,600, 200, and 200 source-document identifiers, respectively. These source-corpus partitions are disjoint by their stored document identifiers; the original test partition is not a human-annotated evaluation set in this study. Tianchi material came from the Recruitment Dataset (ID 163746). The public-institution crawler revisited selected pages weekly or monthly.

The initial annotation collection contains 2,601 records: 150 human-reference sentences and

2,451 original silver records. The silver pool comprises 100 initially reviewed records, 730 additional records from the model-disagreement queue, and 1,621 historical extraction records. The 980-record queue supplied the 150-sentence human reference and the first 830 silver records. Model-specific inclusion rules produce the supervision splits described in Experimental Setup.

The corpus combines four collection batches: AI, Cloud, public-sector, and graduate recruitment. They were obtained through three acquisition routes: MacroData, public-institution websites, and Tianchi. The original training partition contains 10,312 graduate-recruitment sentences and 7,148 sentences from the AI collection; development contains 2,143 sentences from the AI collection; test contains 1,423 from the AI collection, 473 from Tianchi/cloud, and 1,341 from public institutions. The expanded data use three source categories: Tianchi/cloud, public institutions, and listed companies. The blinded sample contains 26, 16, and 8 sentences, respectively.

MacroData's operator is Chongqing Maherui Information Technology Co., Ltd. Public-institution records retain source names, titles, URLs, listed publication dates, and geographic fields where available. Publication dates are not collection dates; a complete crawl-window and source-exclusion history has not been recovered. The authors confirm permission to use and share the MacroData materials for non-commercial academic research. Repository notices distinguish the rights in author-created annotations from the restrictions on third-party texts.

Table 22 gives expanded-pool candidate and retained counts. Intermediate preparation counts are documented in the preparation memo.

Table 22. Candidate and retained record counts in the expanded pools.

Component	Candidates	Adopted pool	Alternative pool
Earlier silver supervision	—	2,077	2,077
First two new batches	3,620	3,407	3,407
Third new batch	1,816	1,703	1,703
Channel-comparison batch	2,500	2,353	2,459
Final pool	—	9,540	9,646

Pool columns report retained records. Earlier supervision combines stages with different candidate sets.

New sentences were selected using requirement-related keywords, a fixed hash-based order for selecting candidates, and quotas for each source. Before annotation, the texts were cleaned for HTML markup and whitespace, split into sentences, and filtered by length. Checks for excluding previously used texts relied on the stored sentences and available pretraining data. No checksum was retained for the 3.2-million-line pretraining file, so the data seen during pretraining cannot be fully reconstructed. Earlier annotations have been preserved; the available records identify the prompts and handbook versions for each documented batch (Appendix A.4).

Split Membership and Document Overlap

The later experiments divide the data by sentence, whereas the original 22,840-sentence corpus was partitioned by its stored document identifiers. For original records, grouping uses the numeric sentence-ID prefix, which matches the stored corpus document identifier. For added records, grouping uses source-file and row identifiers. The audit verifies consistency with stored metadata, not an independent mapping to the original advertisements. The two identifier systems

have not been linked, so source overlap can be checked only within each system. Qwen, XLM-R-large, and ESCOXLM-R use the original silver split, which contains 2,150 records for training and 169 records for development.

Table 23. Shared source-document groups based on stored identifiers.

Split comparison	Shared groups	Records in first split	Records in second split
Silver train / development	1	4	27
Silver train / human reference	59	531	143
Silver development / human reference	0	0	0
Adopted expanded train / validation	185	1,060	208
Adopted expanded train / test	188	1,076	210
Alternative expanded train / validation	186	1,069	210
Alternative expanded train / test	189	1,079	211

Grouping uses numeric record-ID prefixes for original data and source-file/row identifiers for added data; these identify metadata groups, not independently verified original advertisements. The original silver rows use the 2,150/169 split. Counts refer to sentence records in shared groups; records with different identifiers are counted separately even when their text is identical. Split order follows the first column.

We checked for repeated texts across splits by comparing complete texts as stored and after Unicode NFC normalisation, which gives equivalent Unicode character sequences a consistent representation. Neither comparison found matching texts between the original silver training and development sets used by Qwen, XLM-R-large, and ESCOXLM-R, or between training and the human reference.

A broader check used Unicode NFKC normalisation, which also standardises forms such as full-width and half-width characters, then removed whitespace and ignored letter case. This check found no matching complete texts between the original silver training set and the human reference. It found two normalised complete-text keys shared by training and development in the main 2,150/169 split and one shared by training and validation in each expanded pool; these texts already matched after NFKC normalisation alone. Neither expanded pool had training/test matches under the broader check. These checks compare texts without changing stored sentences or annotation offsets; details appear in the text-overlap analysis. Table 23 reports overlap between groups defined by the stored source identifiers.

The silver training set used by Qwen, XLM-R-large, and ESCOXLM-R contains 2,150 records representing 1,910 distinct complete texts. Records with identical text were retained as separate training examples. To assess the effects of this repetition and of shared source-document groups, further experiments would need to compare training with and without repeated texts and evaluate models on sentences from documents excluded from training.

Blinded-sample overlap. The sample-overlap check found no shared sample IDs or matching complete texts after character normalisation between the 50-sentence blinded sample and either the original silver training/development sets (2,150/169) or the human reference set. Of these 50 sentences, each expanded pool contains 46 in training, two in validation, and one in testing; one sentence is absent. The independent annotations support agreement analysis, but these sentences cannot provide an unseen test set for models trained on the expanded pools.

Resource Access and Privacy Screening

Table 24. Research materials and access conditions.

Resource	Contents	Access
Dataset v0.1.3	Corpus, Chinese v4.2.14 handbook (9 September 2026), 150-sentence human reference, and 2,150/169 silver training/development records	Dataset archive
JobBERT-zh	Domain-adapted encoder and CRF; silver-trained checkpoints	Initialisation release; silver-trained release
Qwen models	Three LoRA adapters trained on the original silver data, selected adapters trained on expanded data, predictions, prompts, and scoring code	Model archive; base weights and supervision obtained separately
Expanded silver pools	9,540 adopted and 9,646 alternative records	Public split IDs and selected outputs; complete inputs through the corresponding author for confidential peer review
Historical annotation and assisted review	Source texts, pseudonymised annotator IDs, human annotation layers, source hashes, and sample membership	Annotation archive
Blinded agreement study	50 sentences, three individual annotation layers, source IDs, scorer, and intervals	Agreement release
Chinese-supervised encoders	Six checkpoints, predictions, code, and configurations	Model index and evaluation archive

Scores can be recalculated from the released predictions; rerunning a model also requires its weights and input data. Scores for all six encoder runs were checked locally using saved predictions; results from rerunning inference on the server are documented separately. The five familiarisation and 15 practice sentences are documented separately from the blinded study's 50 formal sentences. The expanded-pool files were checked locally against the recorded split membership. Data Availability describes the reuse terms for each resource; Appendix D reports contact-pattern screening and the limits of documented privacy review.

An automated screen on 4 October 2026 examined 15 selected public corpus and annotation files, covering 28,635 distinct complete texts. Email, mobile-number, and landline patterns flagged 38, 74, and 50 texts, respectively; categories overlap. No such candidates were found in the human reference, original silver development set, or additional review sample. Candidates occur in the source corpus, original silver training set, and separately released blinded agreement sample. The flagged details include organisational recruitment contacts. Recruitment use alone does not establish whether a contact identifies an individual, and some pattern matches may be false positives. The screen does not establish the absence of other personal information. The screened source files were already public. Aggregate counts are also public, while the masked candidate-level review list remains local. The authors report LLM-assisted de-identification and individual checks of human-labelled materials, but the available records do not establish a completed contact-by-contact privacy review of all public corpus and silver-training texts. The original pretraining corpus was unavailable and was not screened.

Limits to training reproduction. The JobBERT-zh fine-tuning experiments used the released Chinese recruitment encoder and its existing CRF layer. The exact original tokeniser version, some pretraining and model-selection settings, and the numbers of inputs truncated or divided into windows were not recovered. The recorded maximum input length for JobBERT-zh was 256 tokeniser tokens. This length is measured after tokenisation; scoring retains all reference spans in the original sentences. The exact Qwen model revision on Hugging Face was not recorded, although file checksums identify the archived files. These missing details limit reproduction of training, but the released predictions can still be used to recalculate scores.

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COMPETING INTERESTS

The authors declare that they have no competing interests.

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